SIMD and GPUs

CS 2410

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Flynn's Taxonomy of Parallelism

SISD Single instruction stream Single data stream Single Core	(SIMD) Single instruction stream Multiple data stream GPUs, Vector Units
MISD Multiple instruction stream Single data stream TPUs	(MIMD) Multiple instruction stream Multiple data stream Multi-core, SMT

SIMD Architectures

ISA not optimized for data parallel workloads

- This loop is called SAXPY (**S**ingle-precision **A X P**lus **Y**):

```
for (int i = 0; i < 64; i++) {
  y[i] = a * x[i] + y[i]
}
```

  - Part of BLAS (**B**asic **L**inear **A**lgebra **S**ubroutines)
  - Frequent in digital signal processing, deep learning, etc.

- Note how we apply the **same operations on each data item**
  - This is how many data parallel workloads look like

- A conventional ISA (likes MIPS) is not optimal for encoding this
  - Results in wasted work and suboptimal performance
  - Let's look at the actual MIPS translation

MIPS code for $y(i) = a * x(i) + y(i)$

```
       l.d    $f0,0($sp)       ;$f0 = a
       addi   $s2,$s0,512      ;64 elements (64*8=512 bytes)
loop:  l.d    $f2,0($s0)       ;$f2 = x(i)
       mul.d  $f2,$f2,$f0      ;$f2 = a * x(i)
       l.d    $f4,0($s1)       ;$f4 = y(i)
       add.d  $f4,$f4,$f2      ;$f4 = a * x(i) + y(i)
       s.d    $f4,0($s1)       ;y(i) = $f4
       addi   $s0,$s0,8        ;increment index to x
       addi   $s1,$s1,8        ;increment index to y
       subu   $t0,$s2,$s0      ;evaluate i < 64 loop condition
       bne    $t0,$zero,loop   ;loop if not done
```

- Blue instructions don't do actual computation. There for indexing and loop control.
  - Is there a way to avoid? Loop unrolling yes. But that causes code bloat!
- Red instructions do computation. But why decode them over and over again?
  - Is there a way to fetch and decode once and apply to all data items?

SIMD (Single Instruction Multiple Data)

- **SIMD (Single Instruction Multiple Data)**
  - An architecture for applying one instruction on multiple data items
  - ISA includes **vector instructions** for doing just that
    - Along with **vector registers** to hold multiple data items

- Using MIPS vector instruction extensions:

```
l.d      $f0,0($sp)    ;$f0 = scalar a
lv       $v1,0($s0)    ;$v1 = vector x (64 values)
mulvs.d  $v2,$v1,$f0   ;$v2 = a * vector x
lv       $v3,0($s1)    ;$v3 = vector y (64 values)
addv.d   $v4,$v2,$v3   ;$v4 = a * vector x + vector y
sv       $v4,0($s1)    ;vector y = $v4
```

  - Note: no indexing and loop control overhead
  - Note: each instruction is fetched and decoded only once

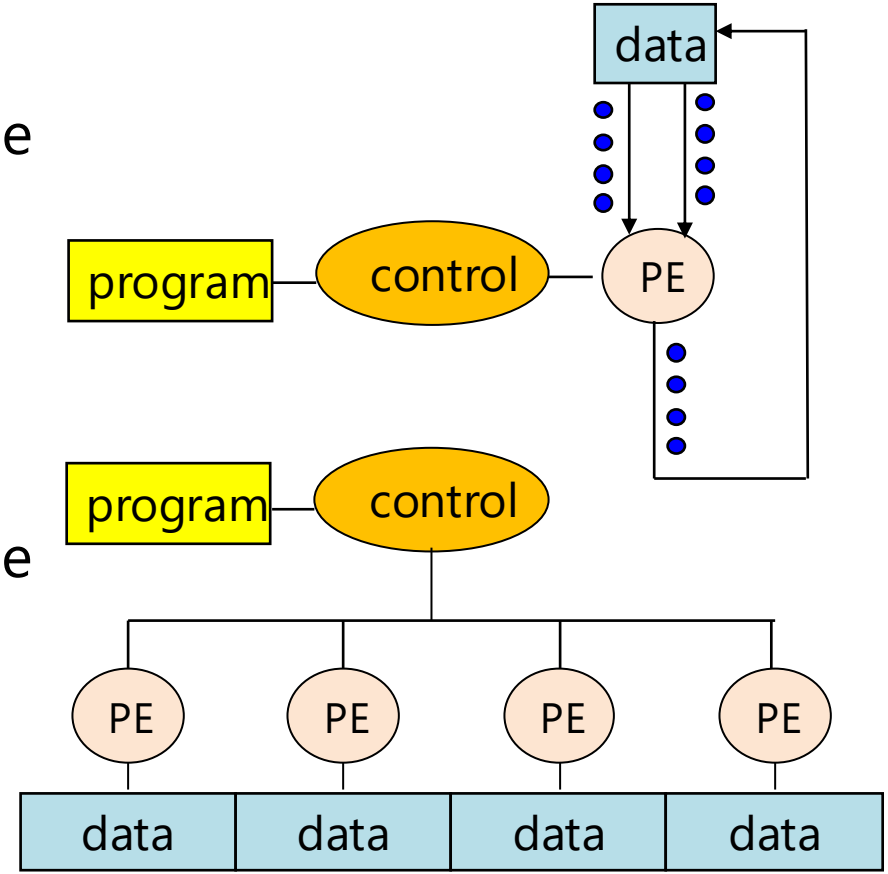
SIMD Processor Design

- How would you design a processor for the vector instructions?

1. One processing element (PE)
   - Fetch and decode instruction once
   - PE applies op on each data item
     - Item may be in vector register
     - Item may be in data memory

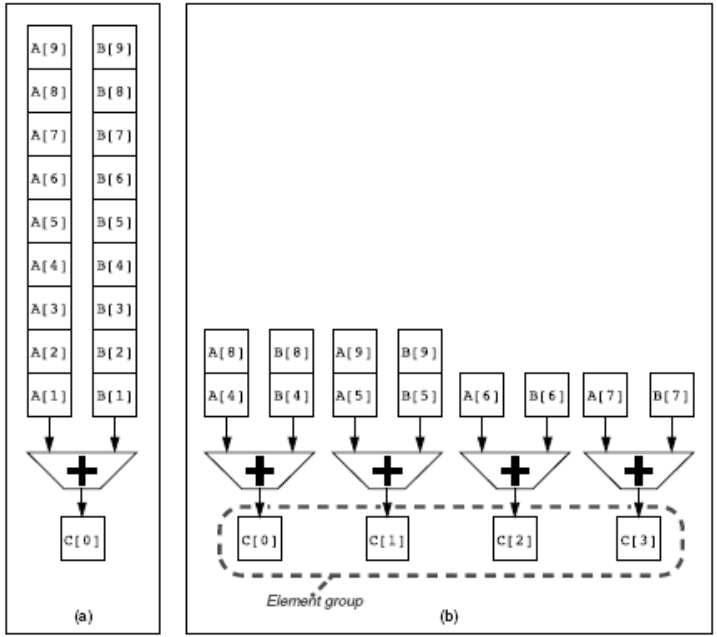
2. Multiple PEs in parallel
   - Fetch and decode instruction once
   - PEs apply op in parallel
     - In synchronous lockstep

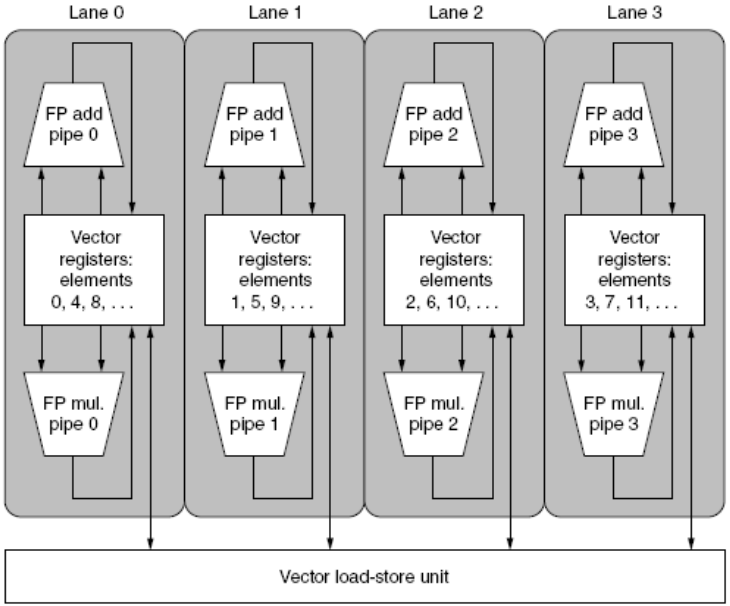
   → The more PEs, the faster!



Vector Unit Design

- Instead of having a single FP adder work on each item (a)
- Have four FP adders work on items in parallel (b)
- Each pipelined FP unit is in charge of pre-designated items in vector
  - For full parallelization, put as many FP units as there are items





Vector Unit Design

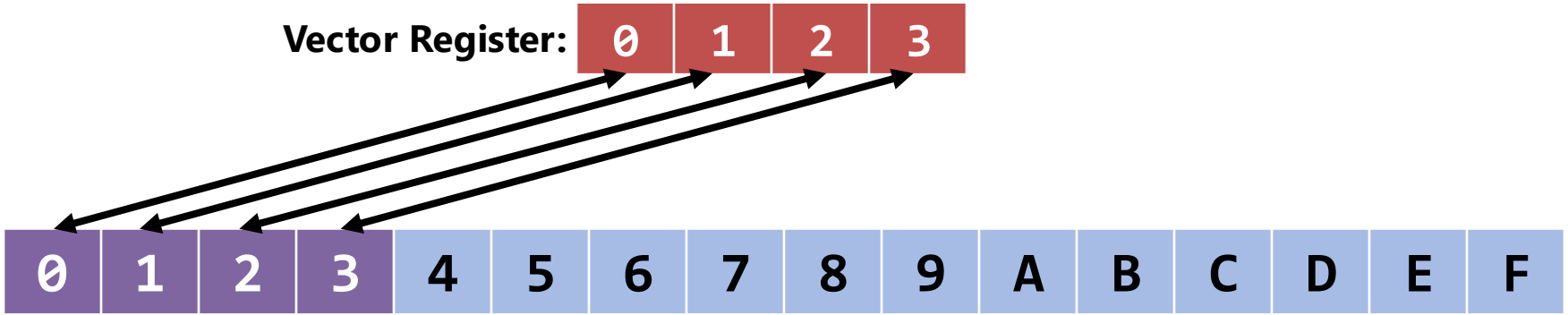
- Why is this better than a 4-wide superscalar with 4 FP ALUs?
- Let's see what's needed to sustain the increased width.

	4-wide Vector Unit	4-wide SuperScalar
Fetch Width	1 instruction / cycle	4 instructions / cycle
Dependency Detection / Dynamic Scheduling	None needed	Need to find 4 independent ALU ops and schedule them
Register File Read Ports	2 ports / lane	8 ports for 4 ALUs
Register File Write Ports	1 port / lane	4 ports for 4 ALUs
Register File Size	Size / lane stays constant	Approx. 4X size for 4 ALUs

- Can efficiently support wide vector units
  - With minimal impact on cycle time
  - With minimal impact on power consumption

Vector Load-Store Unit Design

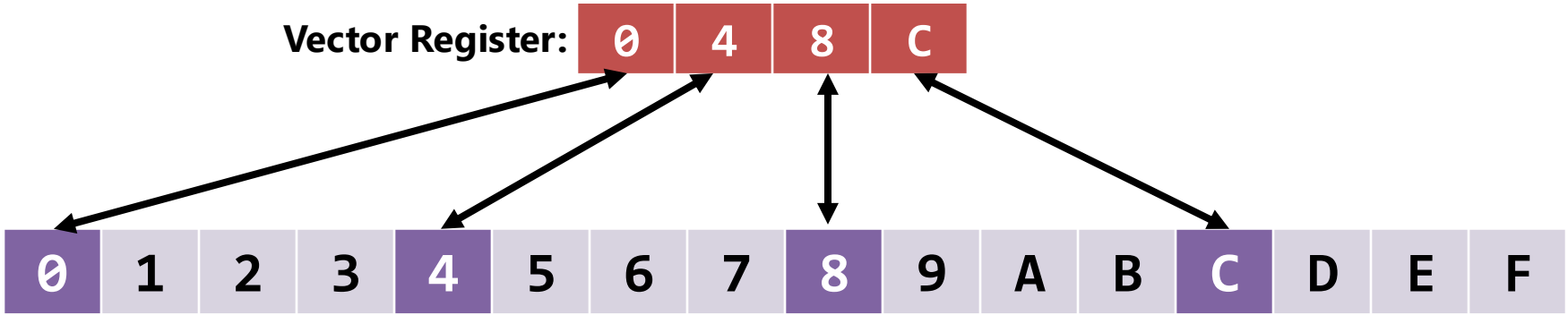
- All vector units allow load/store of *contiguous* data from memory:



- Due to spatial locality, results in transfer of fewest cache blocks
- Best if data is contiguous, but sometimes it is not!

Vector Load-Store Unit Design

- *Striding* lets you load/store *non-contiguous* data from memory at regular offsets. (e.g. the first member of each struct in an array)



- Less spatial locality compared to contiguous data
  → Results in transfer of more cache blocks and more bandwidth
- Hence, AoS (Array of Structs) to SoA (Struct of Arrays) optimization

```
struct point { float x; float y; };
struct point points[N];
```

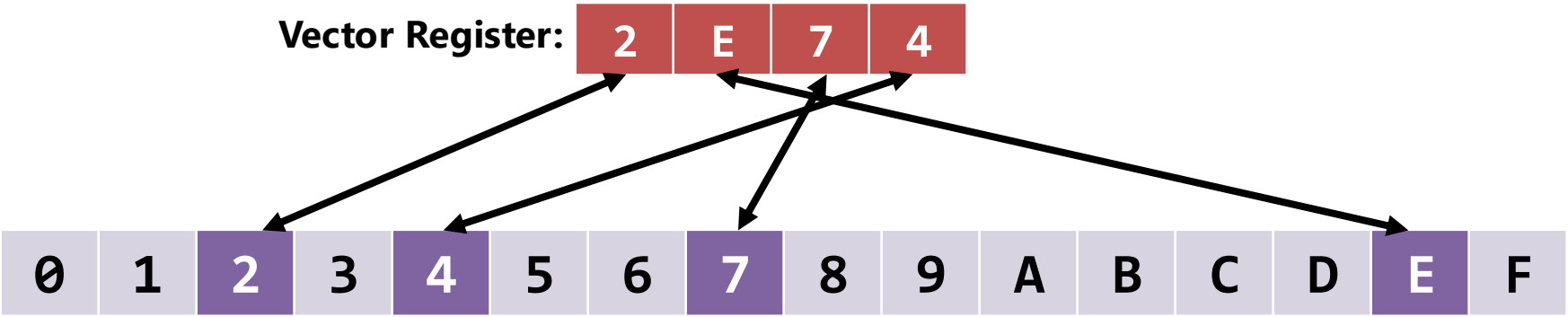
→

```
struct pointArray { float x[N]; float y[N]; };
struct pointArray points;
```

→ Now vector operations on "x" field results in contiguous accesses

Vector Load-Store Unit Design

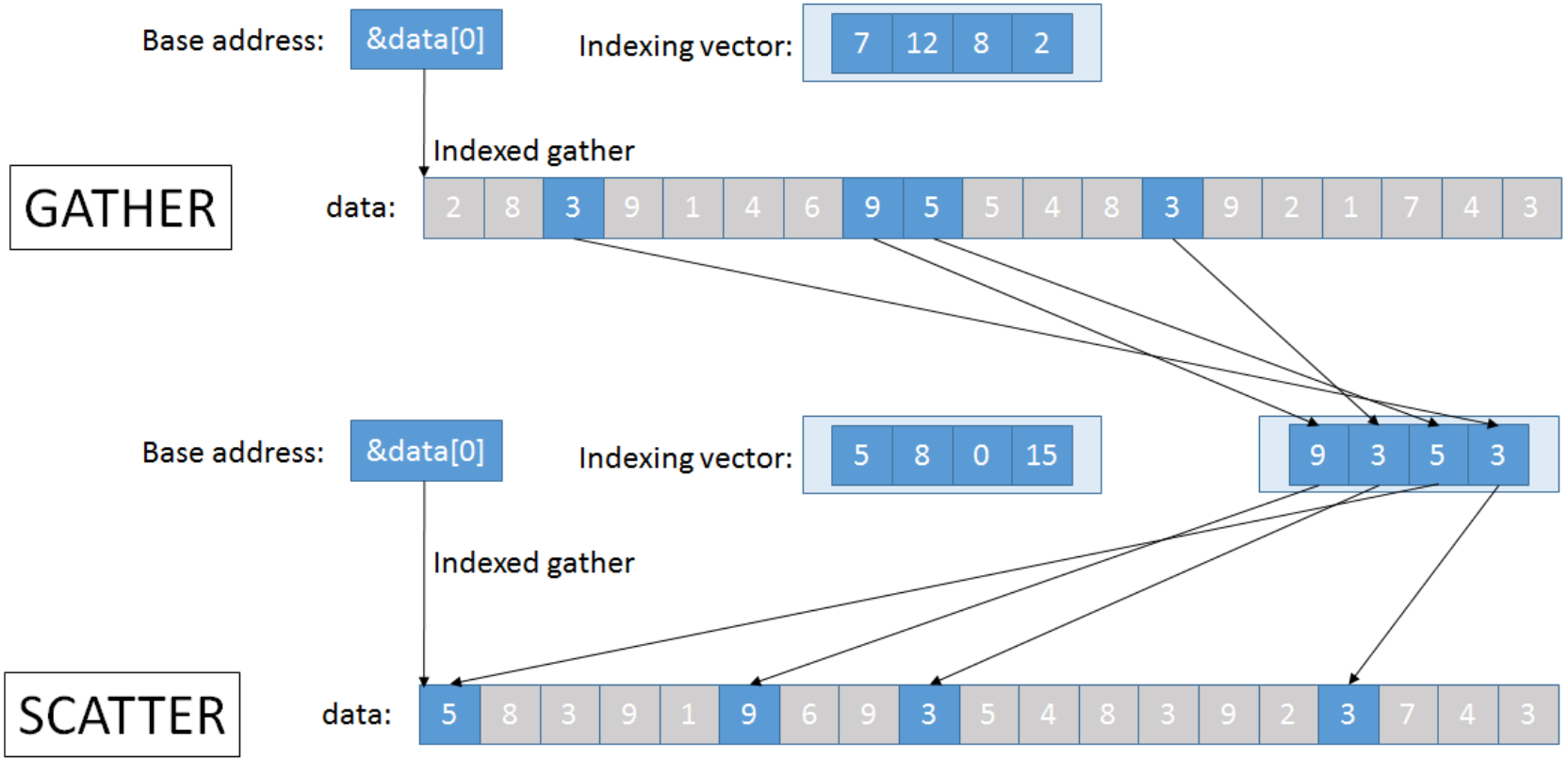
- *Gather-Scatter* lets you put pointers in a vector, then load/store from *arbitrary memory addresses*. *(gather = load, scatter = store)*



- Worst for spatial locality, and hard to do prefetching

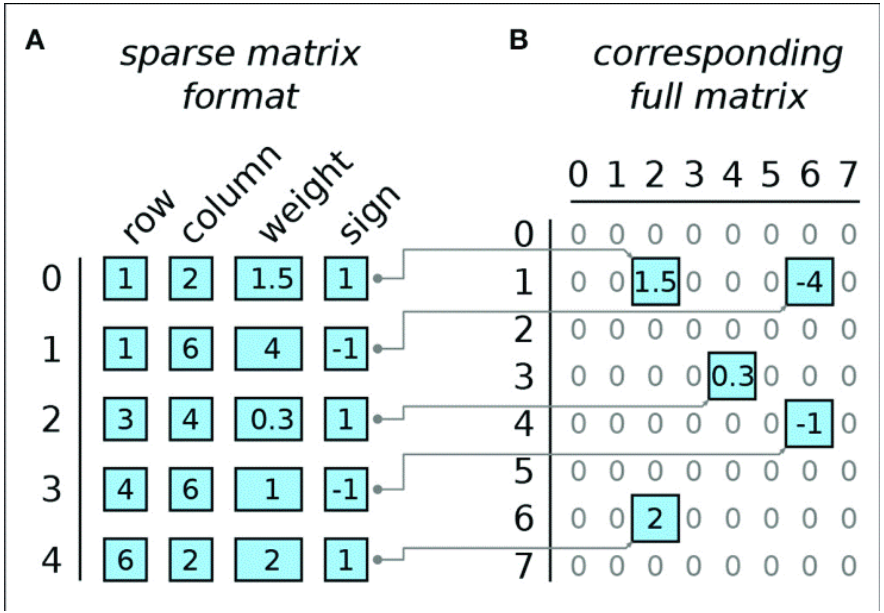
- If you are going to perform this gather-scatter many times
  → Better to sort and compress data items into contiguous array

How does Gather-Scatter work?



When does Gather-Scatter make sense?

- Often data for scientific or AI applications are **sparse**.
  - Time-sampled points where a sensor measurement changes
  - In social networking, connections in a N x N friendship matrix
  - In neural networks, weights in a filter layer that are non-zero
- To save memory, sparse data is stored in sparse format:



How adjacency matrix is stored in a Graph Neural Network (GNN).

Can answer questions such as:
how likely is user 1 to like the movie?

A) Average of movie ratings by all users, weighted by row 1 in matrix.
= adjacency vector ● ratings vector
(Requires gather of non-zero weights)

Vector code is often memory bound

- Contiguous data access is the best for performance
  - Means processor needs to access only one or a few cache blocks
  - Use regular vector loads and stores whenever possible

- Scattered access is sometimes necessary but bad for performance
  - If any of the multiple cache blocks accessed miss, long latency
  - Accessing multiple blocks also consumes a lot of bandwidth

- If your vector code is slow, often it is because it is memory bound
  - Same goes for GPUs which, as we will learn, is a type of SIMD

SIMD instructions in real processors

- x86 vector extensions
  - Historically: MMX (64-bit), SSE (128-bit), AVX (256-bit), AVX-2 (256-bit)
  - Current: AVX-512 (512-bit)
- ARM vector extensions
  - Historically: VFP (Vector Floating Point)
  - Current: Neon (128-bit vector instructions)

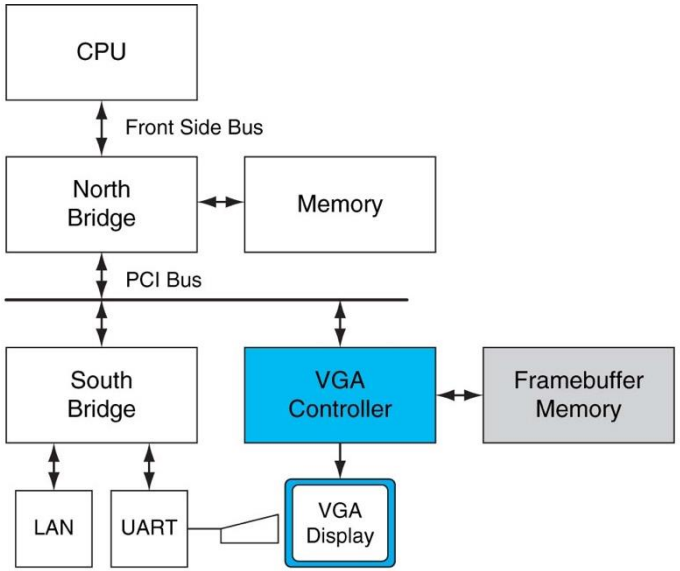
- Vector instructions have progressively become wider historically
  - Due to increase of data parallel applications
  - Due to their power efficiency
    - Compared to fetching, decoding, scheduling multiple instructions
    - Good way to increase FLOPS while staying within TDP limit

- Enter GPUs for general computing (circa 2001)

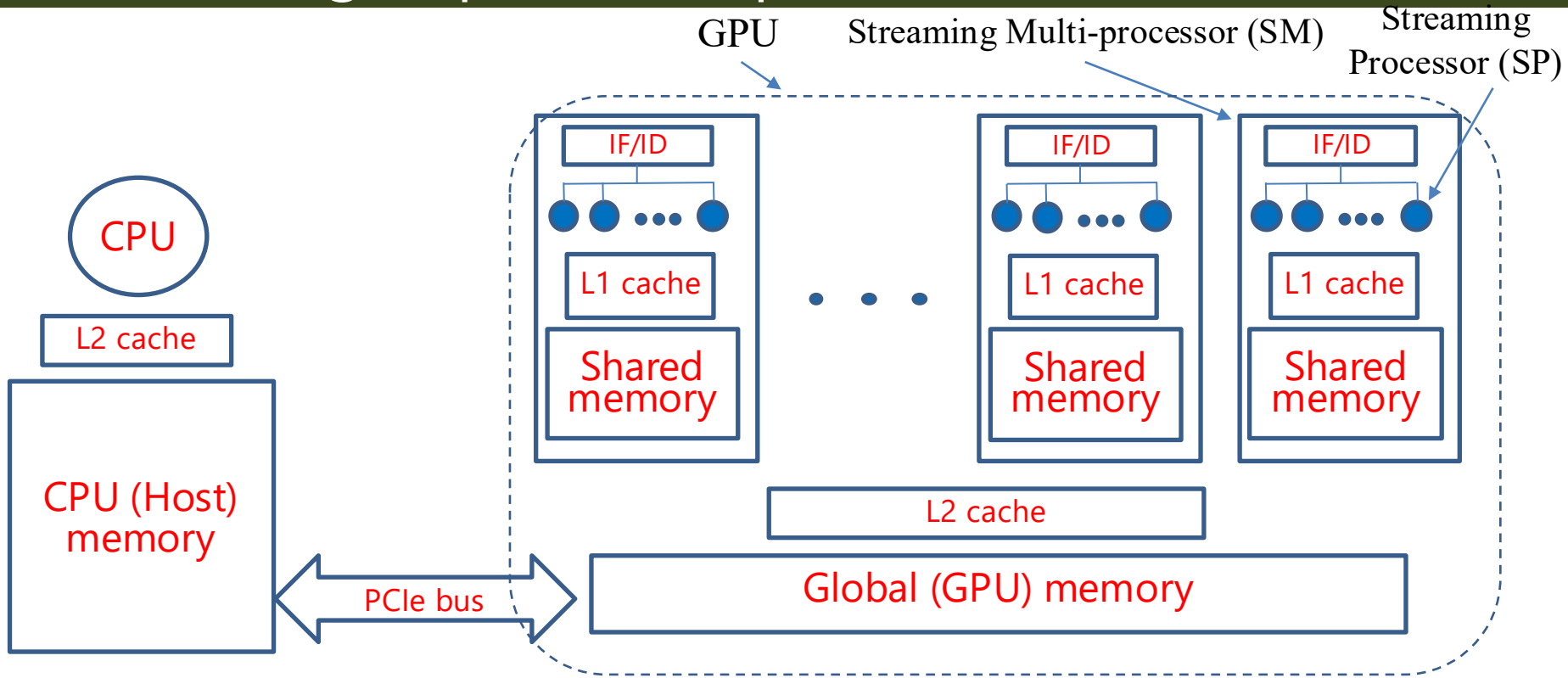
GPUs:
Graphical Processing Units

History of GPUs

- VGA (Video graphic array) has been around since the early 90's
  - A display generator connected to some (video) RAM
- By 2000, VGA controllers were handling almost all graphics computation
  - Programmable through OpenGL, Direct 3D API
  - APIs allowed accelerated vertex/pixel processing:
    - Shading
    - Texture mapping
    - Rasterization
  - Gained moniker Graphical Processing Unit
- 2007: First general purpose use of GPUs
  - 2007: Release of CUDA language
  - 2011: Release of OpenCL language

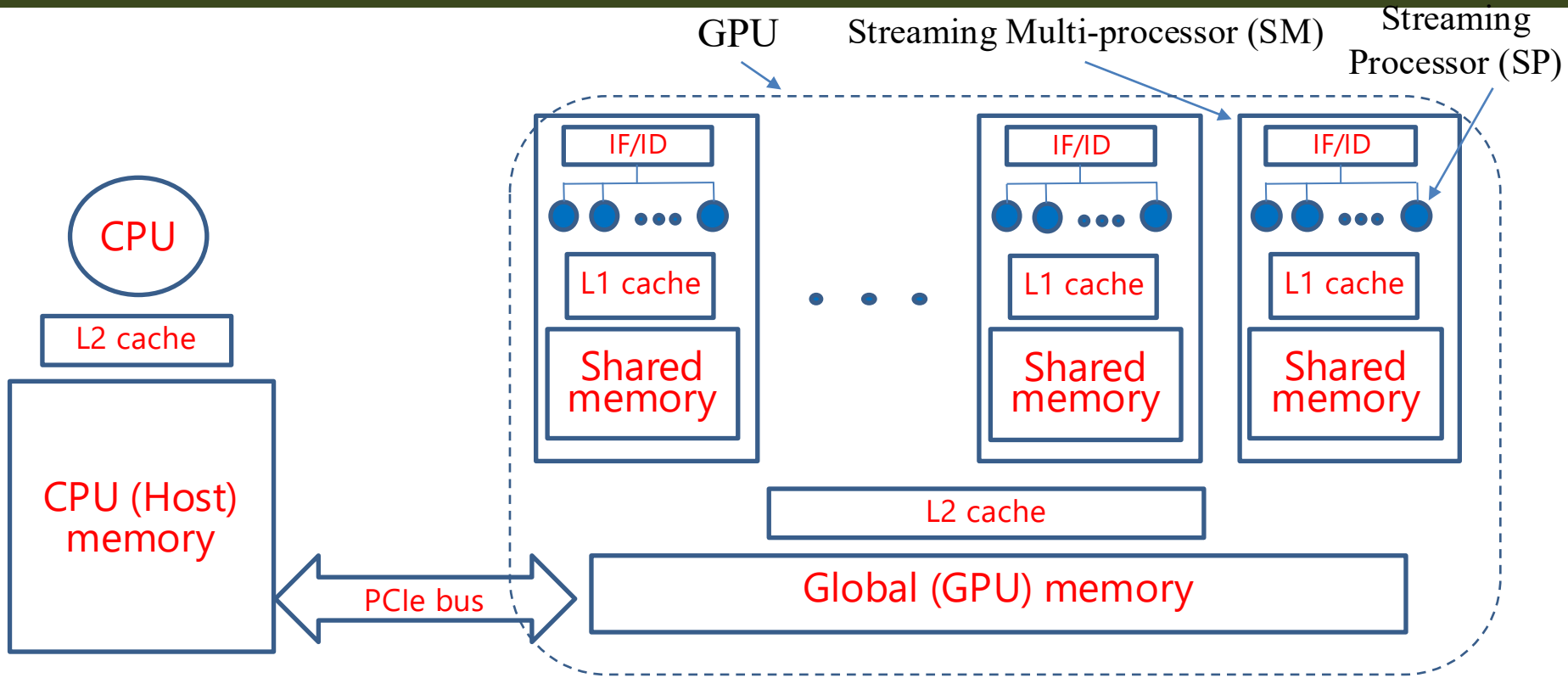


GPU is a group of SIMD processors called SMs



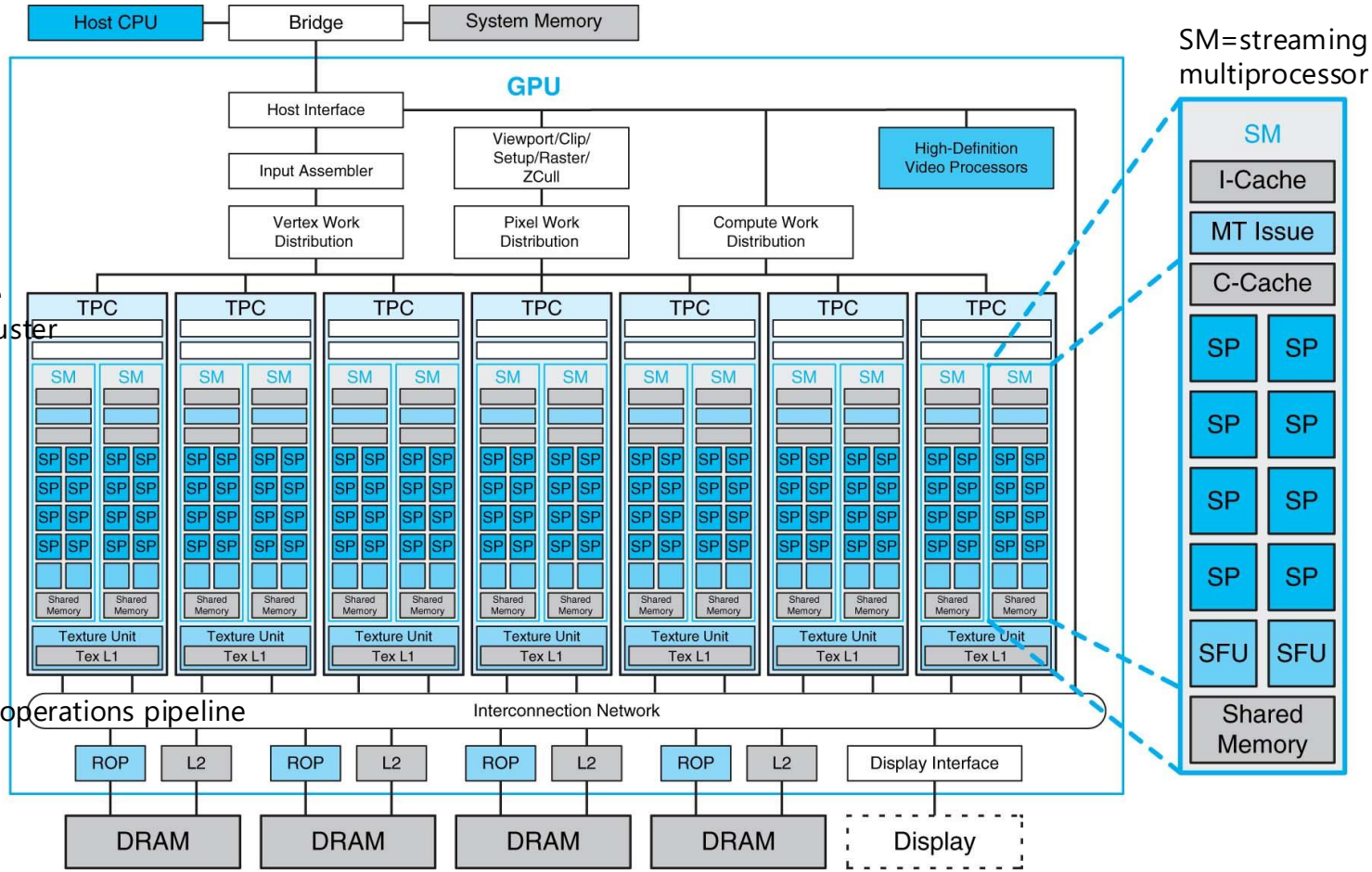
- Logically, a GPU is composed of **SM**s (Streaming Multi-processors)
  - An SM is a **vector** processor that can process multiple pixels (or data items)
- Each SM is composed of **SPs** which work on each pixel or data item

CPU-GPU architecture



- Dedicated GPU memory separate from system memory
- Code and data must be transferred to GPU memory for it to work on it
  - Through PCI-Express bus connecting GPU to CPU

Modern GPU architecture



TPC = texture processing cluster

ROP = raster operations pipeline

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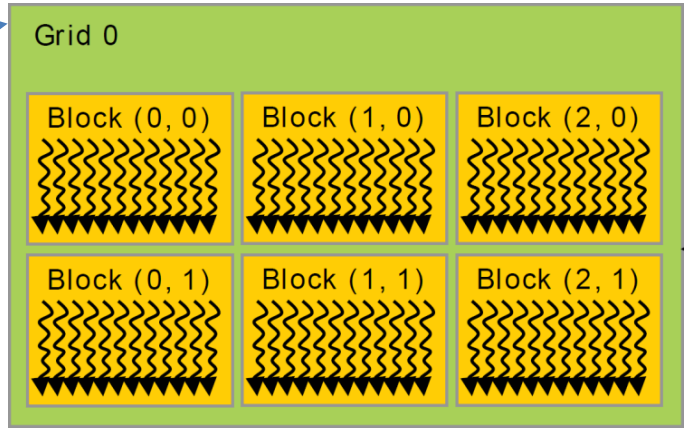
Kernel: A GPU Program

CPU program
(serial code)

```
Function <<<nb,nt >>>
```

Launches kernel with *nb* blocks, each with *nt* threads

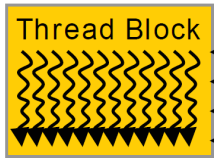


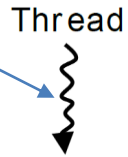
GPU kernel
(SIMD parallel code)

```
_global_ Function ( ... )
```

global designates kernel function (function run by each thread)

Each **thread** works on one data item.





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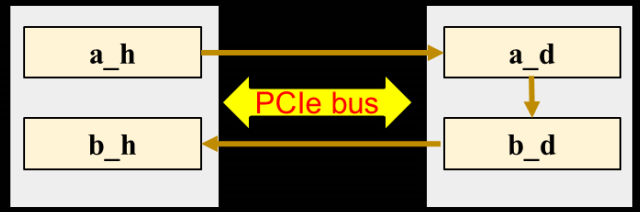
Example: Copying array a to array b using the GPU

Data Movement Example

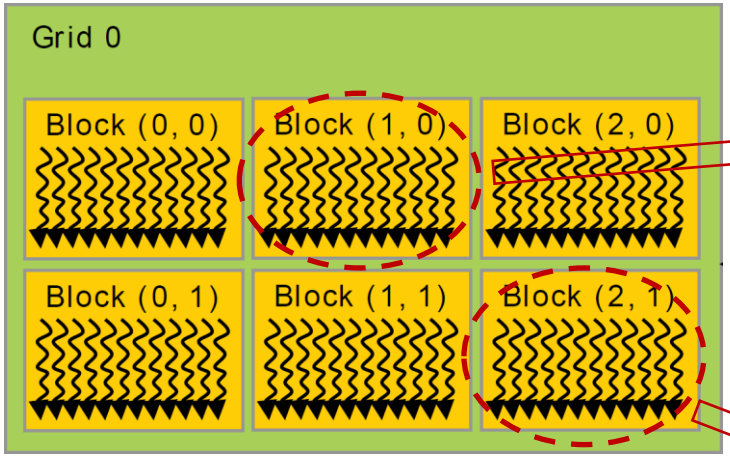
```
int main(void)
{
   float *a_h, *b_h;  // host data
   float *a_d, *b_d;  // device data
   int N = 14, nBytes, i ;

   nBytes = N*sizeof(float);
   a_h = (float *)malloc(nBytes);
   b_h = (float *)malloc(nBytes);
   cudaMalloc((void **) &a_d, nBytes);
   cudaMalloc((void **) &b_d, nBytes);

   for (i=0, i<N; i++) a_h[i] = 100.f + i;

   cudaMemcpy(a_d, a_h, nBytes, cudaMemcpyHostToDevice);
   GPUcomp<<<1, 14>>>(a_d, b_d, N);
   cudaMemcpy(b_h, b_d, nBytes, cudaMemcpyDeviceToHost);

   for (i=0; i< N; i++) assert( a_h[i] == b_h[i] );
   free(a_h); free(b_h); cudaFree(a_d); cudaFree(b_d);
   return 0;
}
```

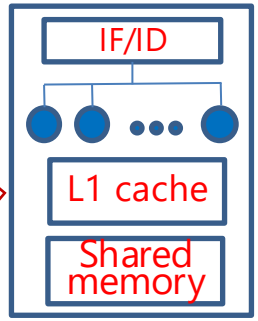



```
_global_ void GPUcomp(*a,*b,N)
{
  int i = threadIdx.x ;
  if( i < N) b(i) = a(i) ;
}
```

The Execution Model of a Kernel

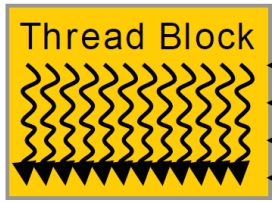
- A **thread block** is the unit of work scheduled to SMs.



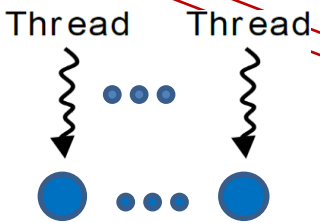


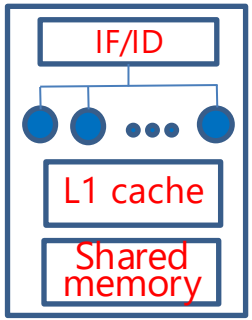
- Each **thread block** is dispatched to an **SM**
- An SM can hold a limited number of thread blocks at a time due to resource constraints (registers, shared memory, ...).

Thread blocks are dispatched to an SM as soon as there is room for another thread block in SM.

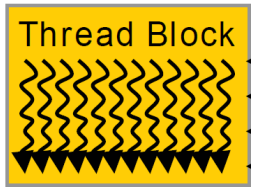


- When a block is dispatched to an SM, each of its threads executes on an SP in the SM.

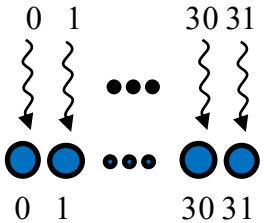




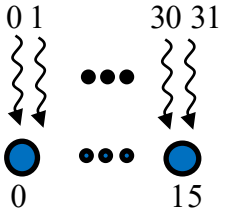
A thread block is executed in warps



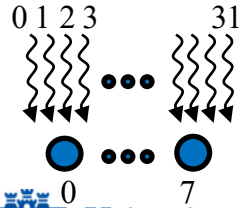
- Each block (up to 1K threads) is divided into groups of 32 threads (called **warps**) – empty threads are used as fillers.
- A warp executes as a SIMD **vector instruction** on the SM.
- Depending on the number of SPs per SM:



  - If 32 SP per warp → 1 thread of a warp executes on 1 SP (32 lanes of execution, one thread per lane)



  - If 16 SP per warp → 2 threads are time multiplexed on 1 SP (16 lanes of execution, 2 threads per lane)



  - If 8 SP per warp → 4 threads are time multiplexed on 1 SP (8 lanes of execution, 4 threads per lane)

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Often an SM has more than one warp!

- In this SM, there are 4 warps.
- Each warp is a vector unit.
  - Executes multiple FP, INT, Tensor ops.
- Warps in thread block are allocated evenly among hardware warps.
- Each warp makes independent progress
  - Has own scheduler and dispatcher.
  - Scheduler? Does that mean it does re-ordering of instructions? No.

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Warps are allocated in Round Robin order

- For a thread block with 32 * 8 = 256 threads:

Warp 0	Warp 1	Warp 2	Warp 3	Warp 4	Warp 5	Warp 6	Warp 7

- For the 4-warp SM we saw in the previous slide:
  - Threads get allocated to hardware warps round robin

Instruction	Warp 0	Warp 1	Warp 2	Warp 3	Warp 4	Warp 5	Warp 6	Warp 7
1								
2								
3								
…								

  - For load balancing, so that each warp gets equal number of threads

Observation: **multiple threads** are allocated on a **single SP**

Each hardware warp is a multi-threaded processor

Instruction	Warp 0	Warp 1	Warp 2	Warp 3	Warp 4	Warp 5	Warp 6	Warp 7
1								
2								
3								
…								

- At each cycle, each warp scheduler chooses threads to execute.
  - **Yellow**: Chose **Warp 0** (out of a choice of 0 and 4)
  - **Green**: Chose **Warp 5** (out of a choice of 1 and 5)
  - **Red**: Chose **Warp 2** (out of a choice of 2 and 6)
  - **Purple**: Chose **Warp 3** (out of a choice of 3 and 7)

Each hardware warp is a multi-threaded processor

Instruction	Warp 0	Warp 1	Warp 2	Warp 3	Warp 4	Warp 5	Warp 6	Warp 7
1								
2								
3								
…								

- At each cycle, each warp scheduler chooses threads to execute.
  - **Yellow**: Chose **Warp 0** (same warp)
  - **Green**: Chose **Warp 5** (same warp)
  - **Red**: Chose **Warp 6** (previously it chose 2)
  - **Purple**: Chose **Warp 7** (previously it chose 3)

Each hardware warp is a multi-threaded processor

Instruction	Warp 0	Warp 1	Warp 2	Warp 3	Warp 4	Warp 5	Warp 6	Warp 7
1								
2								
3								
…								

- Next cycle results in a different set of choices.

Each hardware warp is a multi-threaded processor

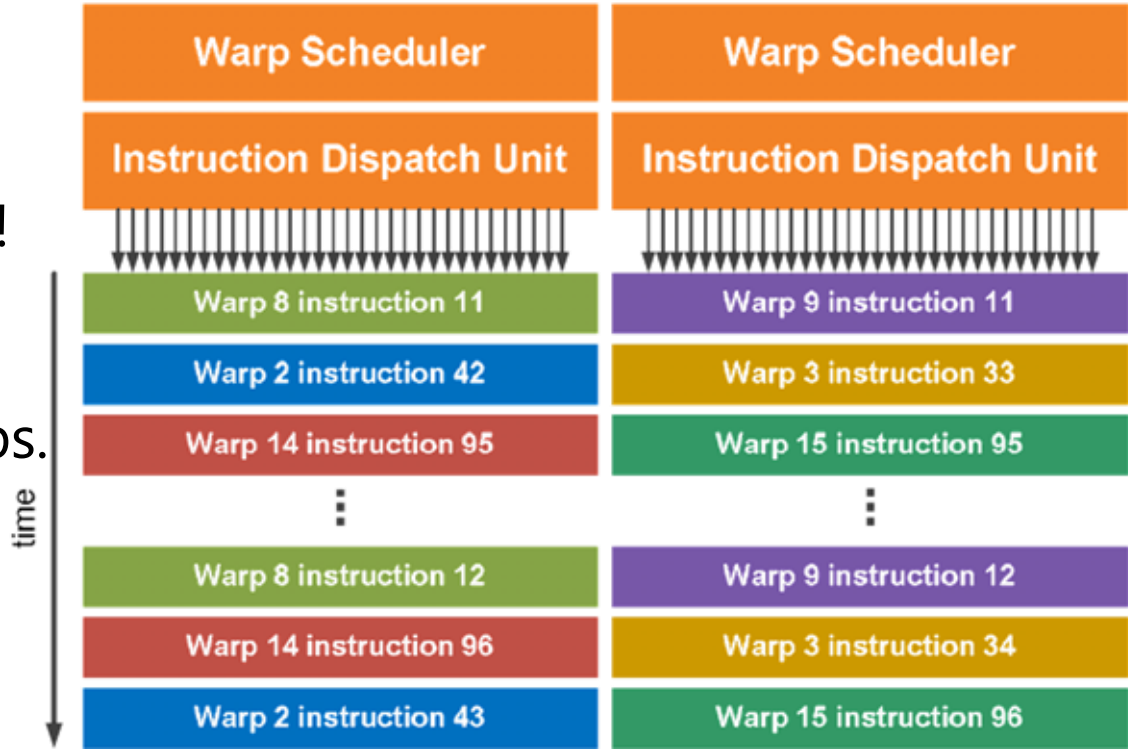
Instruction	Warp 0	Warp 1	Warp 2	Warp 3	Warp 4	Warp 5	Warp 6	Warp 7
1								
2								
3								
…								

- … and a different set of choices for the next cycle.

- What's the point of allowing concurrent progress?
  - Warp scheduler can hide bubbles just like a multi-threaded CPU
- Recall: multi-threading could hide bubbles even for in-order CPUs
  - SPs are also in-order processors, so use the same technique!

Warp scheduling enables latency hiding

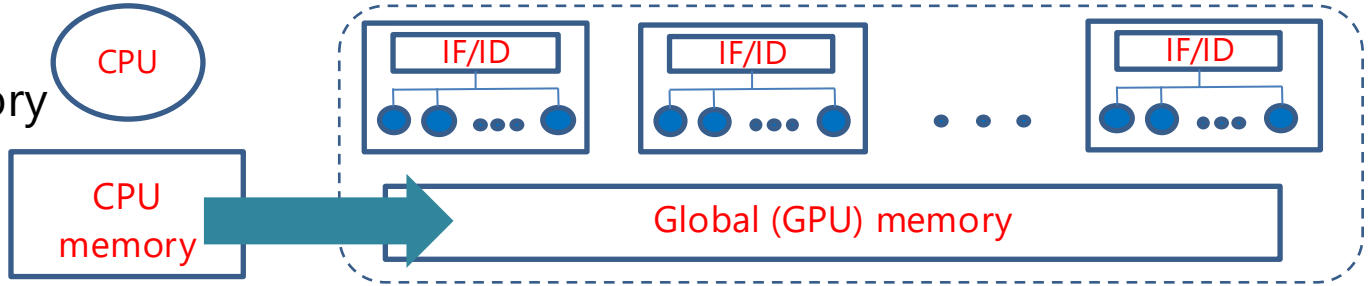
- How?
  - When a warp stalls, instead of idling, switch to a different warp!
- To increase utilization:
  - Have more choice of warps.
  - How?
    - Larger thread blocks
    - More thread blocks



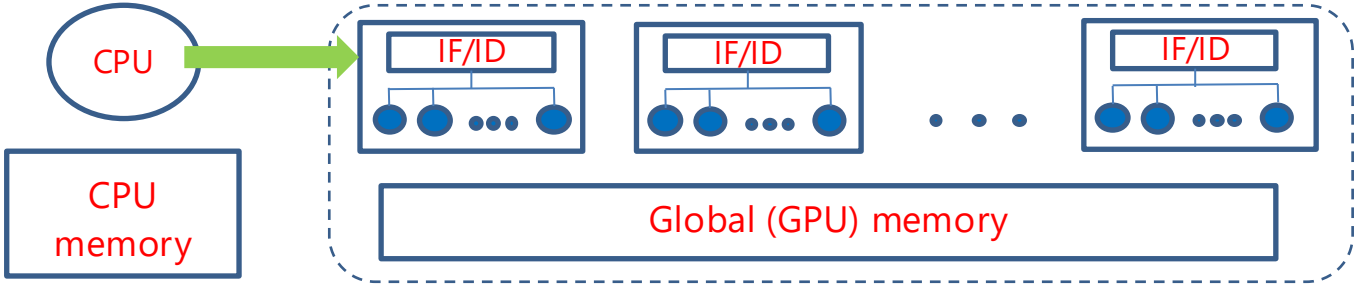
CUDA Programming Model

GPU Programming Model

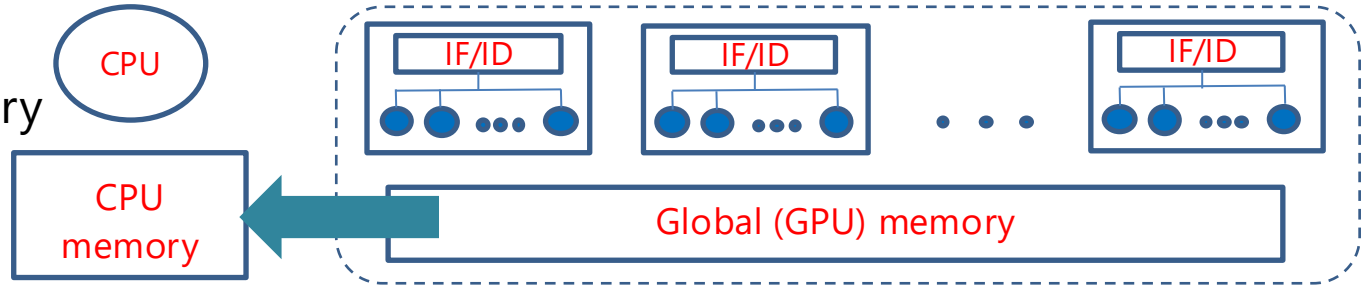
Copy data from CPU memory to GPU memory



Launch the **kernel** (Function executed on the GPU)



Copy data from GPU memory to CPU memory



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GPU Programming Model

CPU program
(serial code)

cudaMemcpy (…)

Copy data from CPU memory to GPU memory

Function <<<nb,nt >>>

Launch kernel on GPU

cudaMemcpy (…)

Copy results from GPU memory to CPU memory

global Function (…)

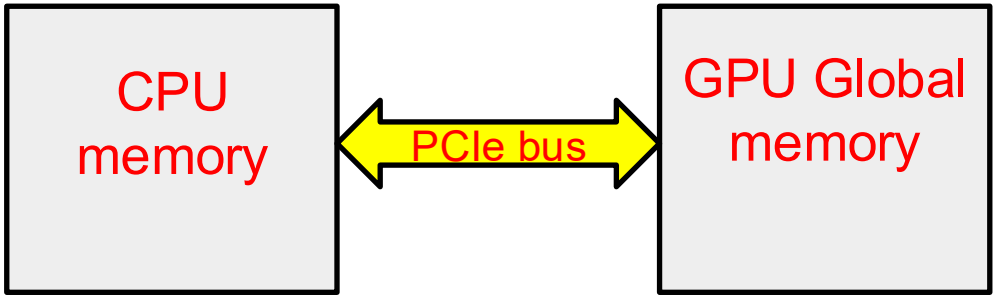
Implementation of GPU kernel
kernel: Function executed on the GPU

GPU Programming Model: Copying Data

```
/* malloc in GPU global memory */
cudaError_t cudaMalloc (void **pointer, size_t nbytes);
/* free malloced GPU global memory */
cudaError_t cudaFree(void *pointer) ;
/* initialize GPU global memory with value */
cudaError_t cudaMemset (void *pointer, int value, size_t count);
/* copy to and from between CPU and GPU memory */
cudaError_t cudaMemcpy (void *dest, void *src, size_t nbytes,
                        enum cudaMemcopyKind dir);
```

enum cudaMemcpyKind

- cudaMemcpyHostToDevice
- cudaMemcpyDeviceToHost
- cudaMemcpyDeviceToDevice



Example: Copying array a to array b using the GPU

Data Movement Example

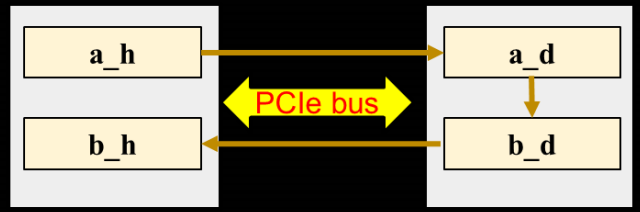
```
int main(void)
{
   float *a_h, *b_h;  // host data
   float *a_d, *b_d;  // device data
   int N = 14, nBytes, i ;

   nBytes = N*sizeof(float);
   a_h = (float *)malloc(nBytes);
   b_h = (float *)malloc(nBytes);
   cudaMalloc((void **) &a_d, nBytes);
   cudaMalloc((void **) &b_d, nBytes);

   for (i=0, i<N; i++) a_h[i] = 100.f + i;

   cudaMemcpy(a_d, a_h, nBytes, cudaMemcpyHostToDevice);
   GPUcomp<<<1, 14>>>(a_d, b_d, N);
   cudaMemcpy(b_h, b_d, nBytes, cudaMemcpyDeviceToHost);

   for (i=0; i< N; i++) assert( a_h[i] == b_h[i] );
   free(a_h); free(b_h); cudaFree(a_d); cudaFree(b_d);
   return 0;
}
```




```
_global_ void GPUcomp(*a,*b,N)
{
  int i = threadIdx.x ;
  if( i < N) b(i) = a(i) ;
}
```

GPU Programming Model: Launching the Kernel

CPU program
(serial code)

cudaMemcpy (…)

Copy data from CPU memory to GPU memory

Function <<<nb,nt >>>

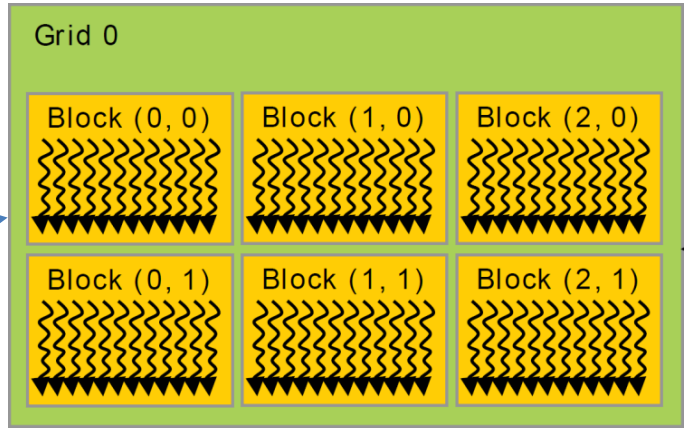
Launch a kernel with *nb* blocks, each with *nt* threads

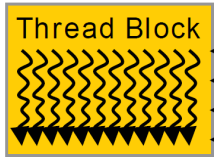
cudaMemcpy (…)

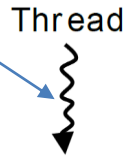
Copy results from GPU memory to CPU memory

global_ Function (…)

Implementation of kernel (the function run by each GPU thread)

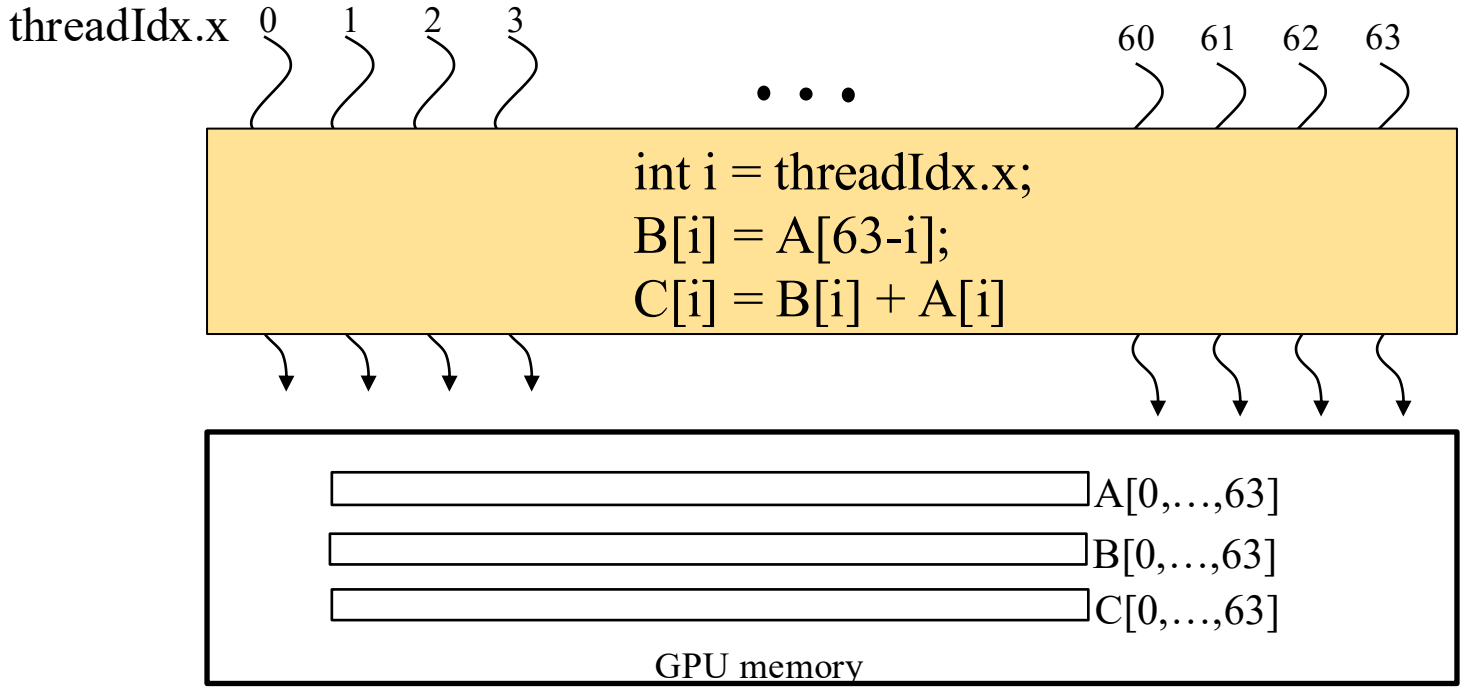






All threads execute the same code

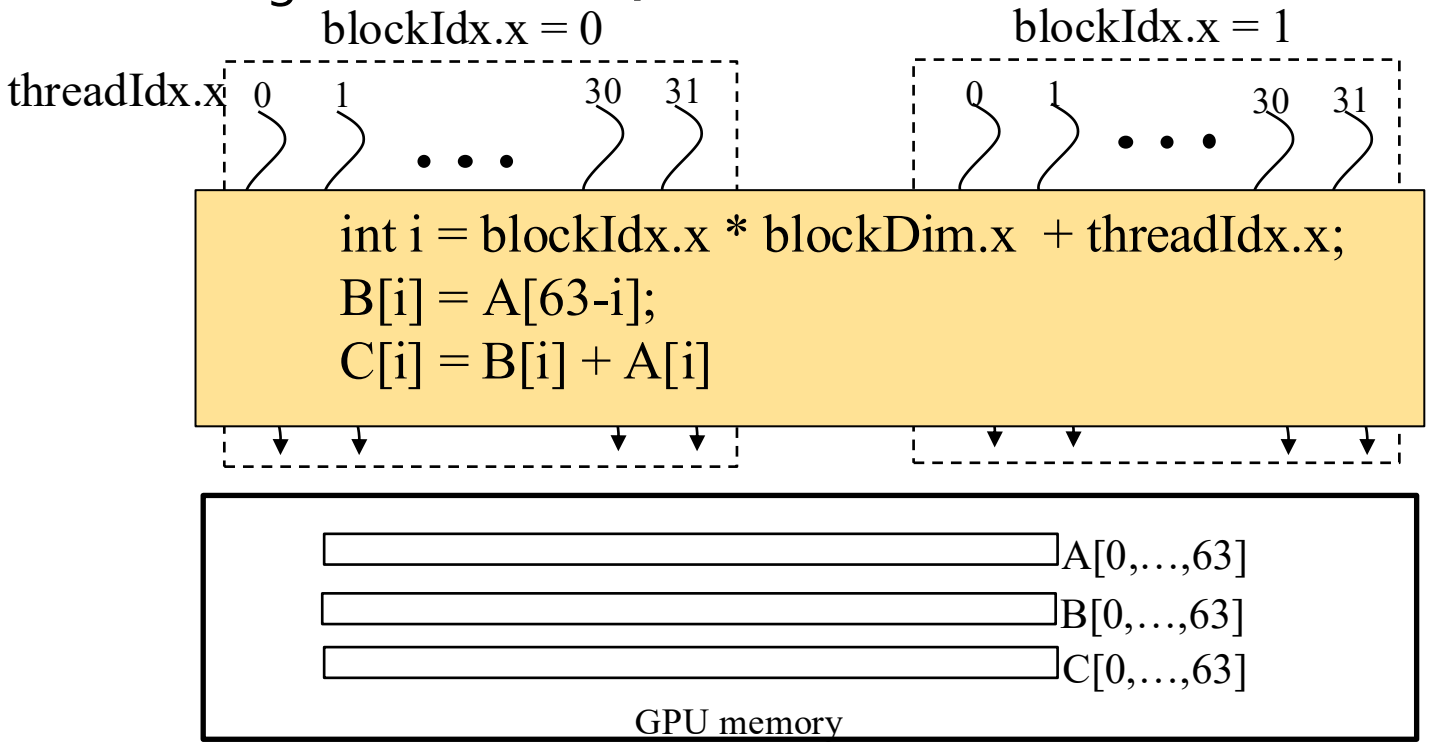
- Launched using **Kernel <<<1, 64>>>** : 1 block with 64 threads



- Each thread in a thread block has a unique "thread index" → **threadIdx.x**
- The same sequence of instructions can apply to different data items.

Blocks of Threads

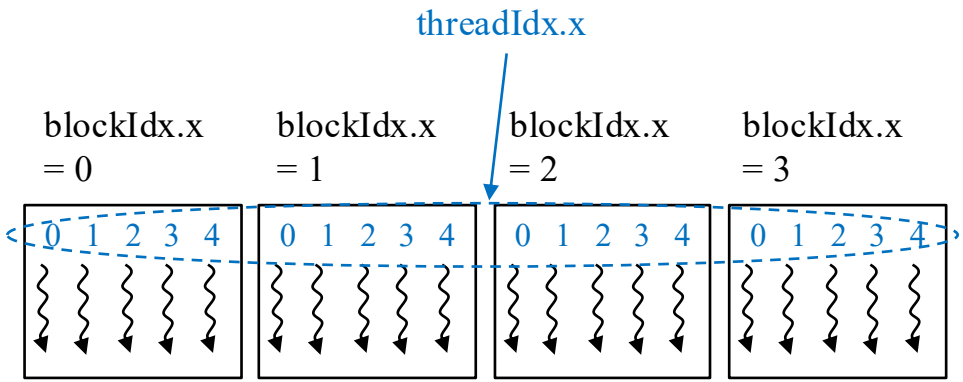
- Launched using **Kernel <<<2, 32>>>** : 2 blocks of 32 threads



- Kernel has a preset thread block size (in this case 32) → **blockDim.x**
- Each thread block has a unique "block index" → **blockIdx.x**
- Can compute index from blockIdx.x, blockDim.x and threadIdx.x

Example: Computing the global index

```
void main ()
{  cudaMalloc (&a, 20*sizeof(int));
   cudaMalloc (&b, 20*sizeof(int));
   cudaMalloc (&c, 20*sizeof(int));
   …
   kernel<<<4,5>>(a, b, c) ;
}
_global_ void kernel(int *a, *b, *c)
{
  int i = blockIdx.x * blockDim.x + threadIdx.x ;
  a[i] = i ;
  b[i] = blockIdx.x;
  c[i] = threadIdx.x;
}
```



NOTE: Each block will consist of one warp – only 5 threads in warp will do useful work. (Other 27 threads will execute no-ops.)

Global Memory

a[]	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
b[]	0	0	0	0	0	1	1	1	1	1	2	2	2	2	2	3	3	3	3	3
c[]	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4

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Example: Computing $y(i) = a * x(i) + y(i)$

C program (on CPU)

```
void saxpy_serial(int n, float a, float
*x, float *y)
{
  for(int i = 0; i<n; i++)
        y[i] = a * x[i] + y[i];
}
```

```
void main ()
{
  …
  saxpy_serial(n, 2.0, x, y);
  …
}
```

CUDA program (on CPU+GPU)

```
_global_ void saxpy_gpu(int n, float a, float *x,
float *y)
{
  int i = blockIdx.x*blockDim.x +
          threadIdx.x;
  if (i < n ) y[i] = a * x[i] + y[i];
}
```

```
void main ()
{ …
  // cudaMalloc arrays X and Y
  // cudaMemcpy data from x, y to X, Y
  int NB = (n + 255) / 256;
  saxpy_gpu<<<NB, 256>>>(n, 2.0, X, Y);
  // cudaMemcpy data from Y to y
}
```

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Example: Computing $y(i) = a * x(i) + y(i)$

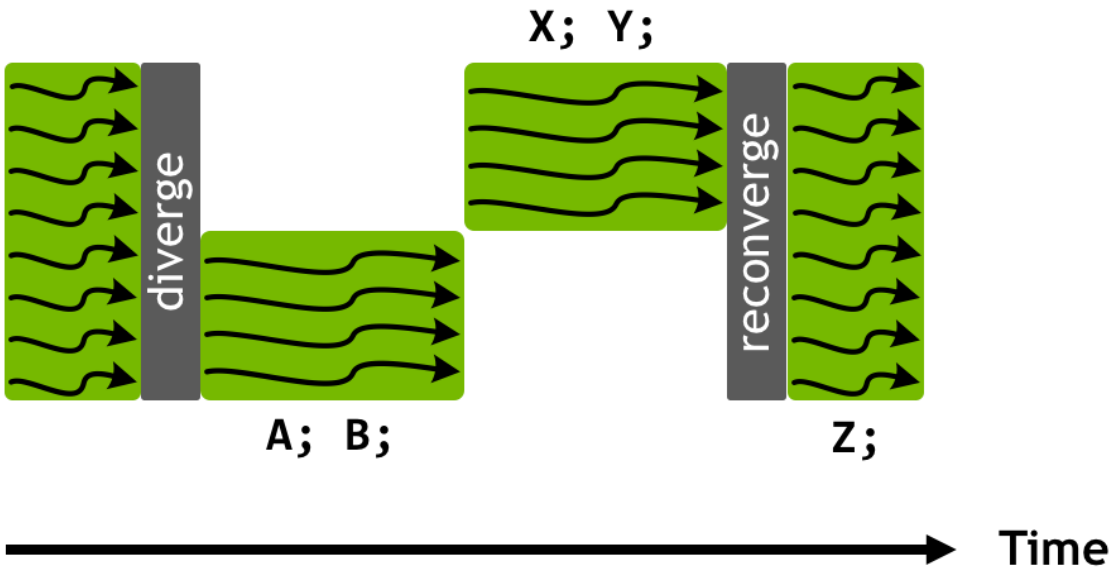
- What happens when n = 1?

```
_global_void saxpy_gpu(int n, float a, float *X, float *Y)
{
  int i = blockIdx.x*blockDim.x +  threadIdx.x;
  if (i < n ) Y[i] = a * X[i] + Y[i];
}
…..
saxpy_gpu<<<1, 256>>>(1, 2.0, X, Y); /* X and Y are both sized 1! */
```

- "if (i < n)" condition prevents writing beyond bounds of array.
- But that requires some threads within a **warp** not performing the write.
  - But a warp is a single vector instruction. How can you branch?
  - "if (i < n)" creates a **predicate** "mask" vector to use for the write
  - Only thread 0 has predicate turned on, rest has predicate turned off

GPUs Use Predication for Branches

```
if (threadIdx.x < 4) {
    A;
    B;
} else {
    X;
    Y;
}
Z;
```

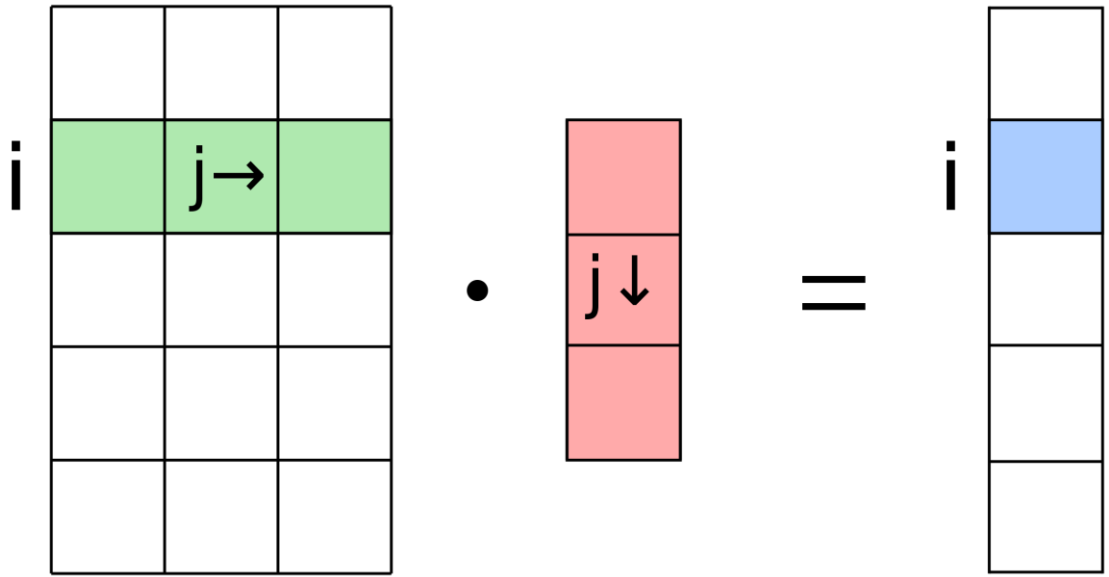


- Each thread computes own predicate for condition threadIdx.x < 4
- Taken together, 32 threads of a warp create a 32-bit predicate mask
- Mask is applied to warps for A, B, X, and Y.
- Just like for VLIW processors, this can lead to **low utilization**.

Example: Computing $y(i) = A(i,j) * x(j)$

- Matrix-Vector Multiply:

```
for (int i=0; i<5; i++)
  for (int j=0; j<3; j++)
    y[i] += A[i*3 + j] * x[j];
```



Example: Computing $y(i) = A(i,j) * x(j)$

C program (on CPU)

```
void mv_cpu(float* y, float* A,
float* x, int n) {
   for (int i=0; i<n; i++)
      for (int j=0; j<n; j++)
         y[i] += A[i*n + j] * x[j];
}
```

```
void main ()
{
   …
   mv_cpu(y, A, x, n);
   …
}
```

CUDA program (on CPU+GPU)

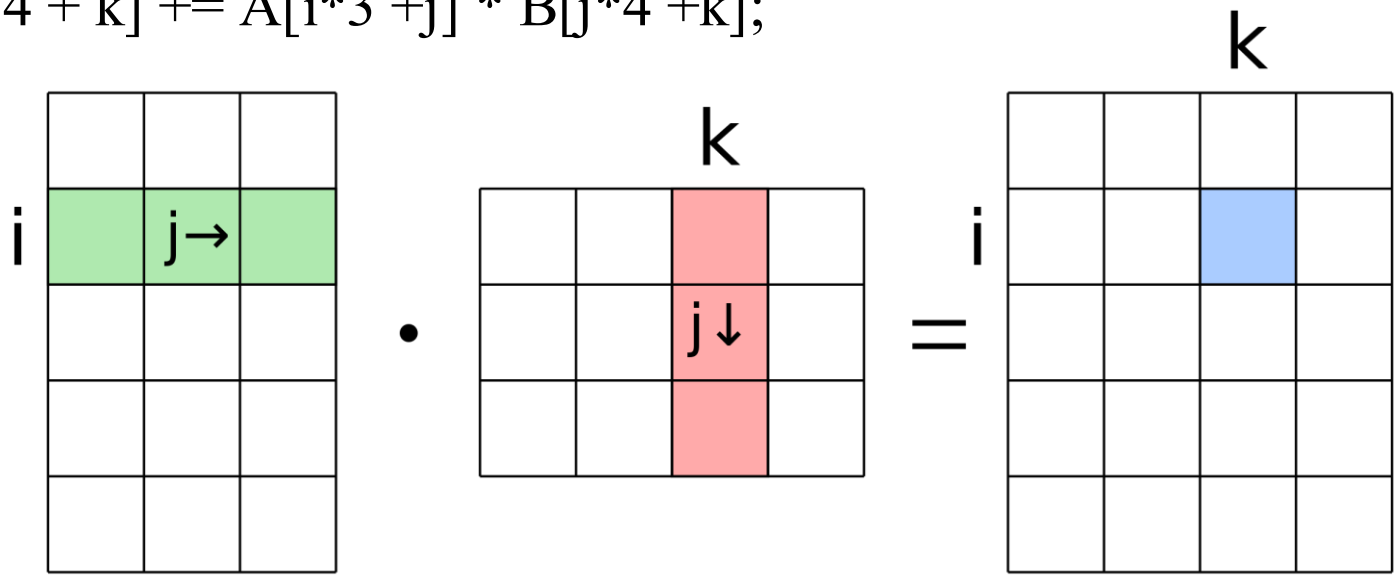
```
void mv_gpu(float* y, float* A, float* x, int n) {
   int i = blockIdx.x * blockDim.x + threadIdx.x;
   if (i < n) {
      for (int j = 0; j < n; j++)
         y[i] += A[i*n + j] * x[j];
   }
}
```

```
void main ()
{
   …
   int nblocks = (n + block_size - 1) / block_size;
   mv_gpu <<<nblocks, block_size>>> (y, A, x, n);
   …
}
```

Example: Computing $C(i,j) = A(i,k) * B(k,j)$

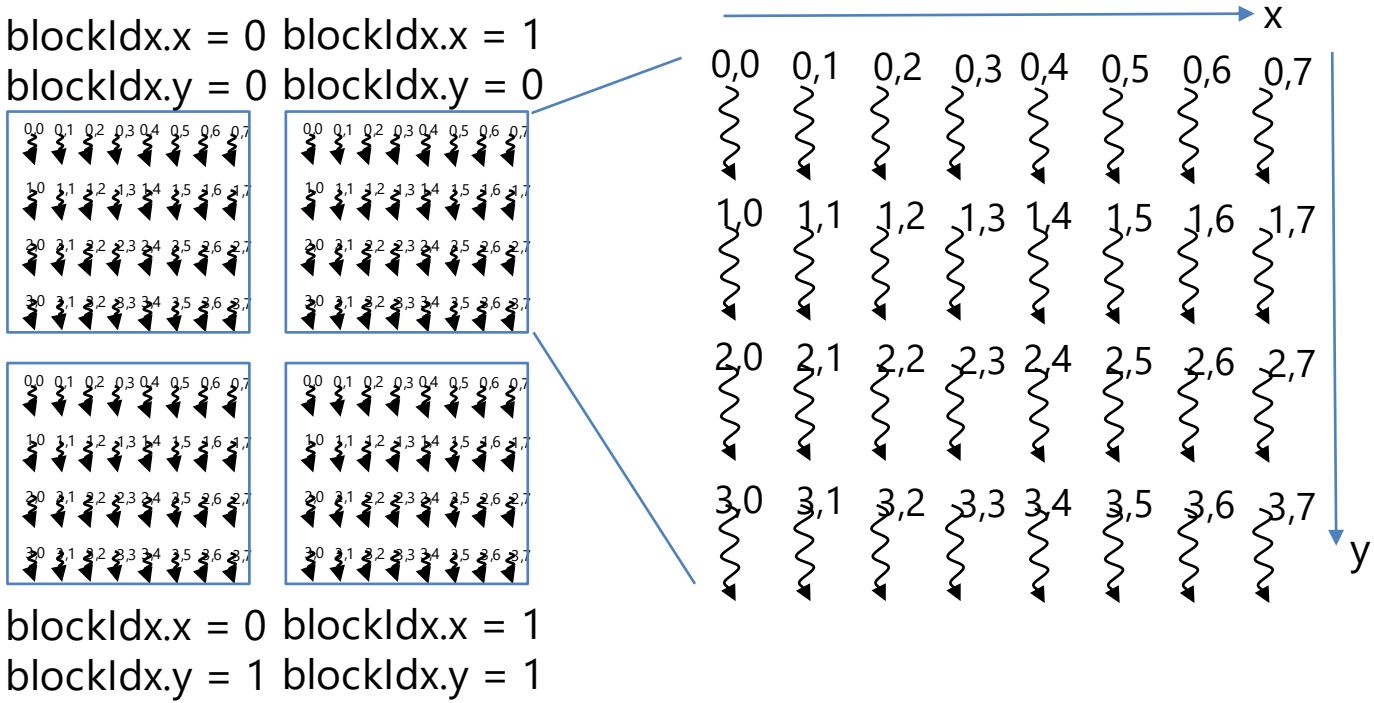
- Matrix-Matrix Multiply:

```
for (int i=0; i<5; i++)
  for (int k=0; k<4; j++)
    for (int j=0; j<3; j++)
      C[i*4 + k] += A[i*3 +j] * B[j*4 +k];
```



Two-dimensions grids and blocks

- Launched using **Kernel <<<(2, 2), (4, 8)>>>** : 2X2 blocks of 4X8 threads



- Each block has two indices **(blockIdx.x, blockIdx.y)**
- Each thread in a thread block has two indices **(threadIdx.x, threadIdx.y)**

Example: Computing $C(i,j) = A(i,k) * B(k,j)$

C program (on CPU)

```
void mm_cpu(float* C, float* A, float* B,
int n) {
  for (int i=0; i<n; i++)
    for (int j=0; j<n; j++)
      for (int k=0; k<n; k++)
        C[i*n + j] += A[i*n + k] * B[k*n + j];
}
```

```
void main ()
{
  mm_cpu(C, A, B, n);
}
```

CUDA program (on CPU+GPU)

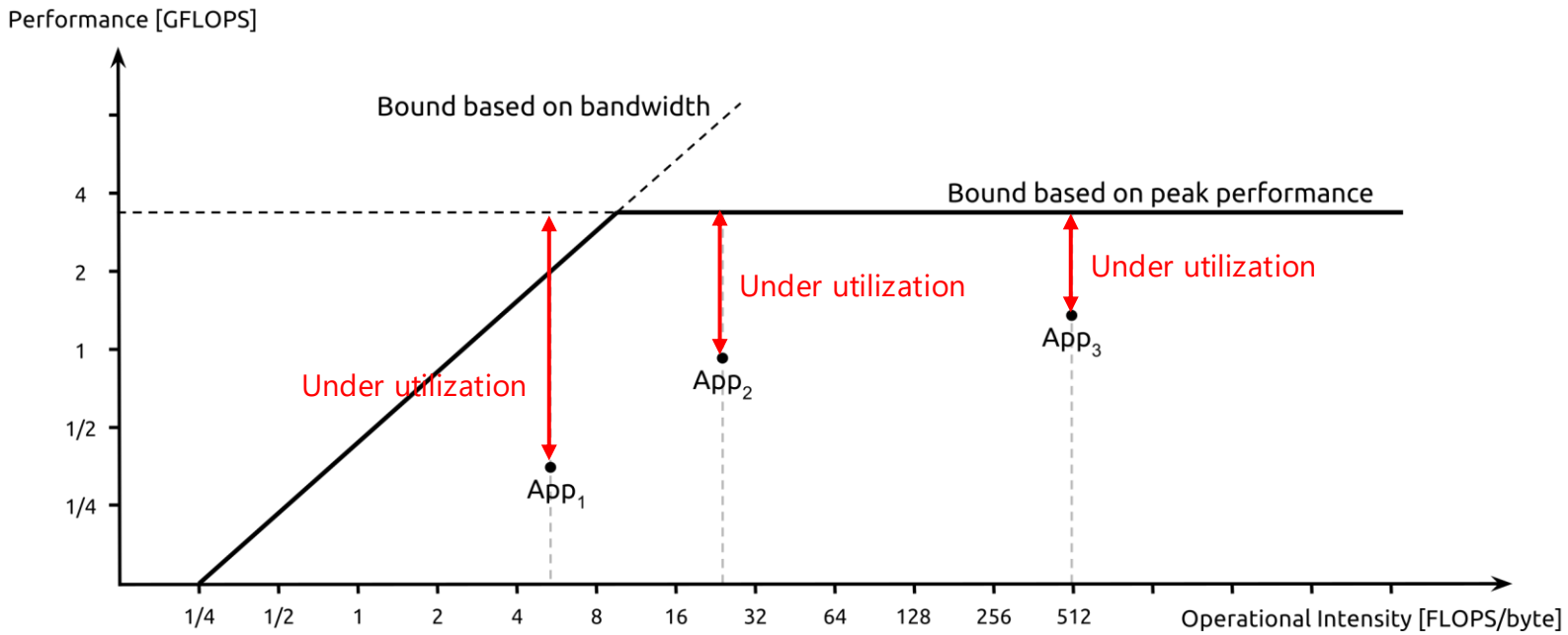
```
void mm_gpu(float* C, float* A, float* B, int n) {
  float Cvalue = 0;
  int i = blockIdx.y * blockDim.y + threadIdx.y;
  int j = blockIdx.x * blockDim.x + threadIdx.x;
  for (int k = 0; k < n; ++k)
    Cvalue += A[i * n + k] * B[k * n + j];
  C[i * n + j] = Cvalue;
}
```

```
void main ()
{
  dim3 dimBlock(block_size, block_size);
  dim3 dimGrid(n / dimBlock.x, n / dimBlock.y);
  mm_gpu <<<dimGrid, dimBlock>>> (C, A, B, n);
}
```

GPU Performance

Roofline Model can be applied to GPUs too

- What should I do if my app is not reaching peak performance?
  1. Ensure app has enough parallelism to fully utilize GPU
  2. Measure memory bandwidth to see if app is memory bound
     - If so, think of ways to improve operational intensity



Lesson 1: Parallelism is important for utilization

Thread Level Parallelism

- Superscalars and VLIWs are useful only if…
  - Program exhibits ILP (Instruction Level Parallelism) in the code
- GPUs are useful only if…
  - Program has **TLP (Thread Level Parallelism)** in the code
  - TLP can be expressed as the number of threads in the code

- How that TLP is laid out in the kernel is important:
  - How many **threads** are in a thread block
    - If less than number_of_warps * 32, some **SPs** may get unused
  - How many **thread blocks** are in the grid
    - If less than number_of_SMs, some **SM**s may get unused
  - Oversubscribe more threads / thread blocks than strictly necessary
    - This helps in latency hiding through warp switching

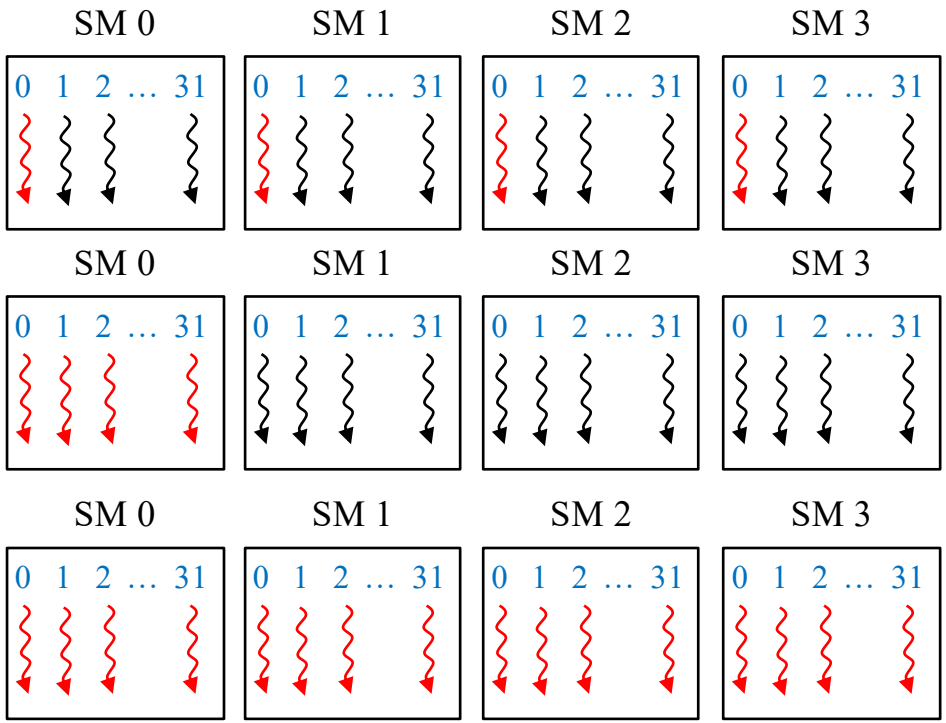
Example: Kernels with Bad Layout

- Suppose there are 4 SMs in GPU with 32 SPs in each SM.
  - Utilized threads are marked in red. Rest are unused.

- Case 1: Not enough threads
  kernel<<<1024, 1>>>(…) ;

- Case 2: Not enough blocks
  kernel<<<1, 1024>>>(…) ;

- Balanced threads and blocks
  kernel<<<32, 32>>>(…) ;
  kernel<<<16, 64>>>(…) ;
  kernel<<<4, 256>>>(…) ;



Lesson 2: Bandwidth is important if memory bound

Example: Computing $y(i) = A(i,j) * x(j)$

C program (on CPU)

```
void mv_cpu(float* y, float* A,
float* x, int n) {
   for (int i=0; i<n; i++)
      for (int j=0; j<n; j++)
         y[i] += A[i*n + j] * x[j];
}
```

```
void main ()
{
   …
   mv_cpu(y, A, x, n);
   …
}
```

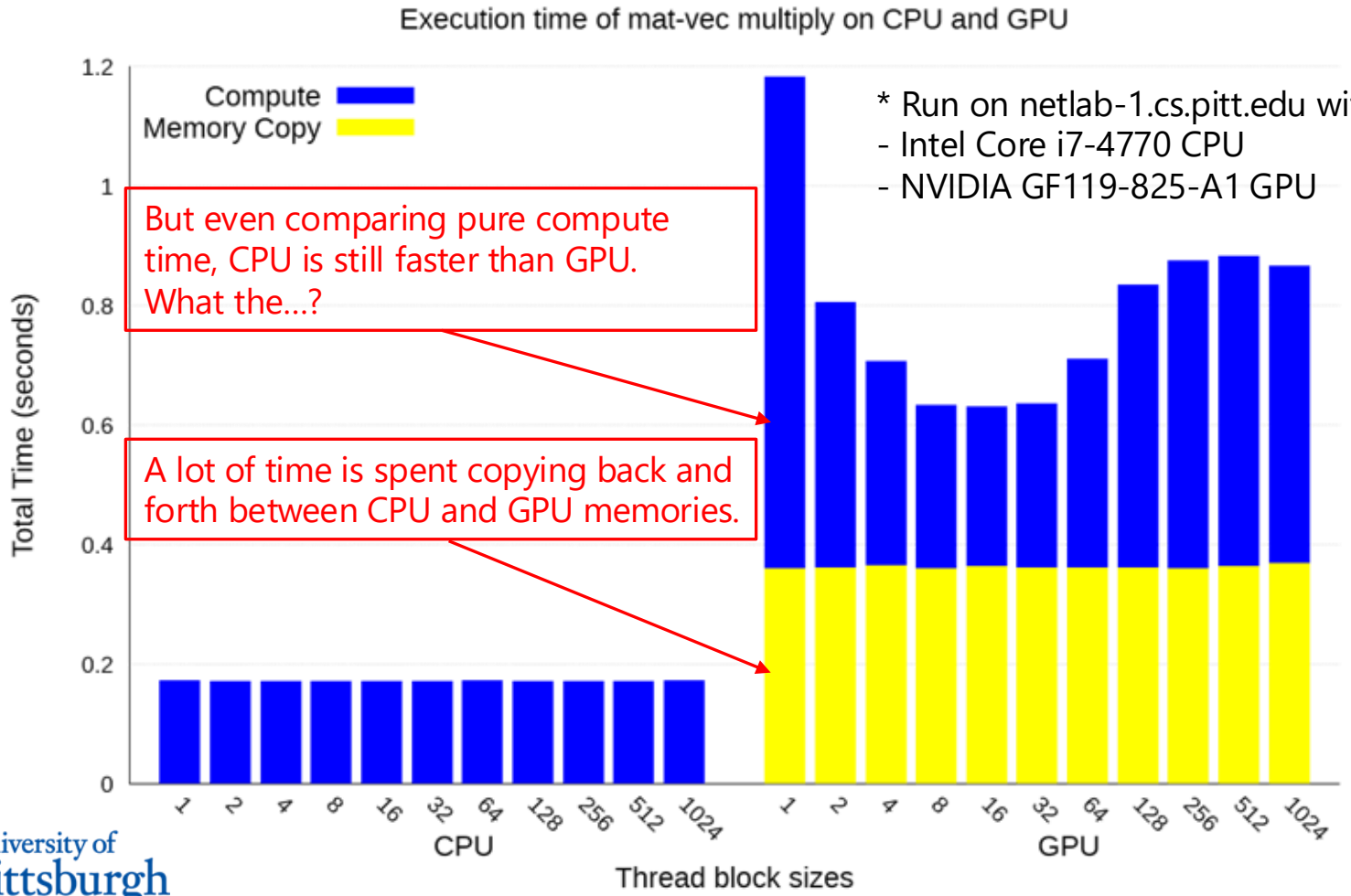
CUDA program (on CPU+GPU)

```
void mv_gpu(float* y, float* A, float* x, int n) {
   int i = blockIdx.x * blockDim.x + threadIdx.x;
   if (i < n) {
      for (int j = 0; j < n; j++)
         y[i] += A[i*n + j] * x[j];
   }
}
```

```
void main ()
{
   …
   int nblocks = (n + block_size - 1) / block_size;
   mv_gpu <<<nblocks, block_size>>> (y, A, x, n);
   …
}
```

Performance Results for $y(i) = A(i,j) * x(j)$

- Guess what? CPU is faster than GPU!



But even comparing pure compute time, CPU is still faster than GPU. What the…?

A lot of time is spent copying back and forth between CPU and GPU memories.

* Run on netlab-1.cs.pitt.edu with n=8192:
- Intel Core i7-4770 CPU
- NVIDIA GF119-825-A1 GPU

Performance Results for $y(i) = A(i,j) * x(j)$

- Was it because the GPU was wimpy and can't do enough **FLOPS**?

- NVIDIA GF119-825-A1 is a Fermi GPU Capability 2.1
  - Clock rate: 1046 MHz (X 2 for warp execution)
  - Number of SMs: 1
  - Number of SPs per SM: 48
  - Max FLOPS = 1046 MHz * 2 * 1 * 48 = **100.4 GFLOPS**

- What was the FLOPS achieved?
  - y[i] += A[i*n + j] * x[j] = 2 FP ops each iteration for n * n iterations
  - n = 8192, so FP ops = 8192 * 8192 * 2 = 134 M
  - Time = 0.27 seconds (shortest at 32 thread block size)
  - FLOPS = 134 M / 0.27 = **496 MFLOPS**
  - **Not even close to the limit!**

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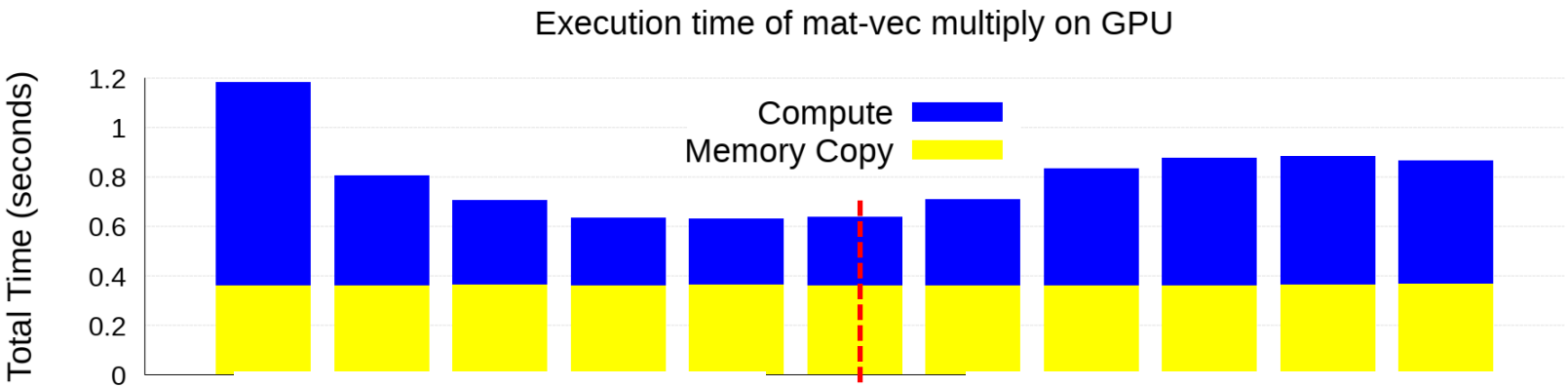
Performance Results for $y(i) = A(i,j) * x(j)$

- Could it be that the GPU didn't have enough **memory bandwidth**?

- NVIDIA GF119-825-A1 is a Fermi GPU Capability 2.1
  - Memory Type: DDR3
  - Memory Bandwidth: **14.00 GB/s**

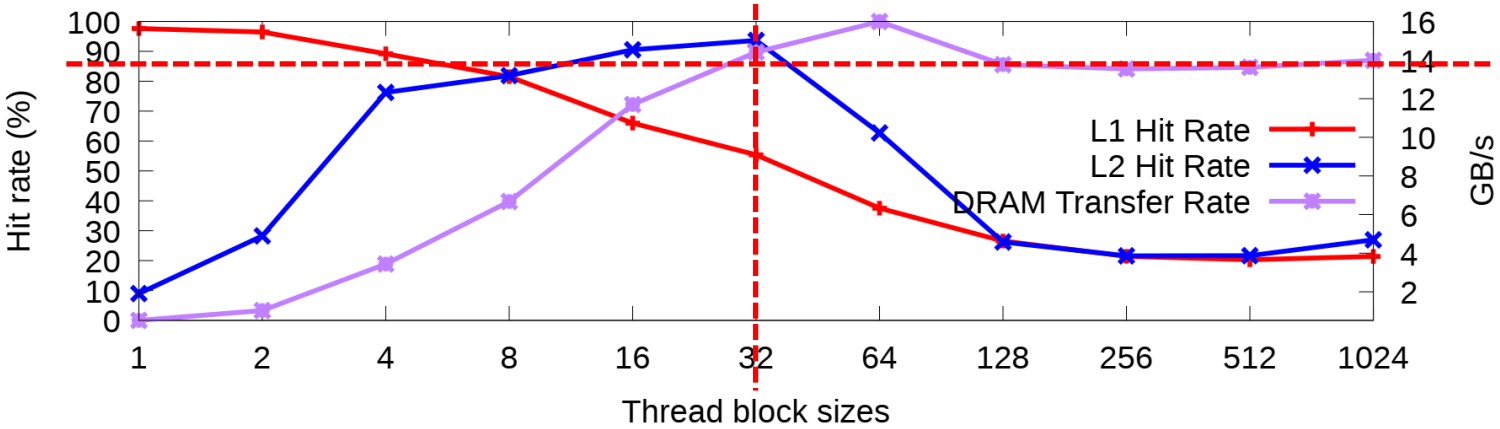
- GPUs also have Performance Monitoring Units (PMUs)
  - NVIDIA Profiler (nvprof) provides an easy way to read them: https://docs.nvidia.com/cuda/profiler-users-guide/index.html
  - Let's use the PMU to profile the following:
    - DRAM Transfer Rate (GB/s)
    - L1 Hit Rate (%)
    - L2 Hit Rate (%)

Memory Wall Hits Again



DRAM bandwidth not saturated:
Benefits from more parallelism
due to larger thread block sizes

DRAM bandwidth saturated:
Larger thread blocks → increased working set size →
higher cache miss rates → worse bandwidth problem



DRAM Bandwidth Limit (14 GB/s)

Is there a way we can reach max FLOPS?

- Let's take a look at the GPU design metrics again:
  - Max FLOPS = 100.4 GFLOPS
  - Memory Bandwidth: 14.00 GB/s

- To sustain max FLOPS, you need to do a lot of work per byte
  - 100.4 GFLOPS / 14.00 GB/s = 7.17 FP ops / byte
  - Or, about **28 FP ops / float** (4 bytes) fetched from memory
  - Otherwise, the memory bandwidth cannot sustain the FLOPS

- All GPUs have this problem with **memory bandwidth**:
  - It's easy to put in more SMs using transistors for Moore's Law
  - Your memory bandwidth is limited due to your DDR interface

Operational Intensity of $y(i) = A(i,j) * x(j)$

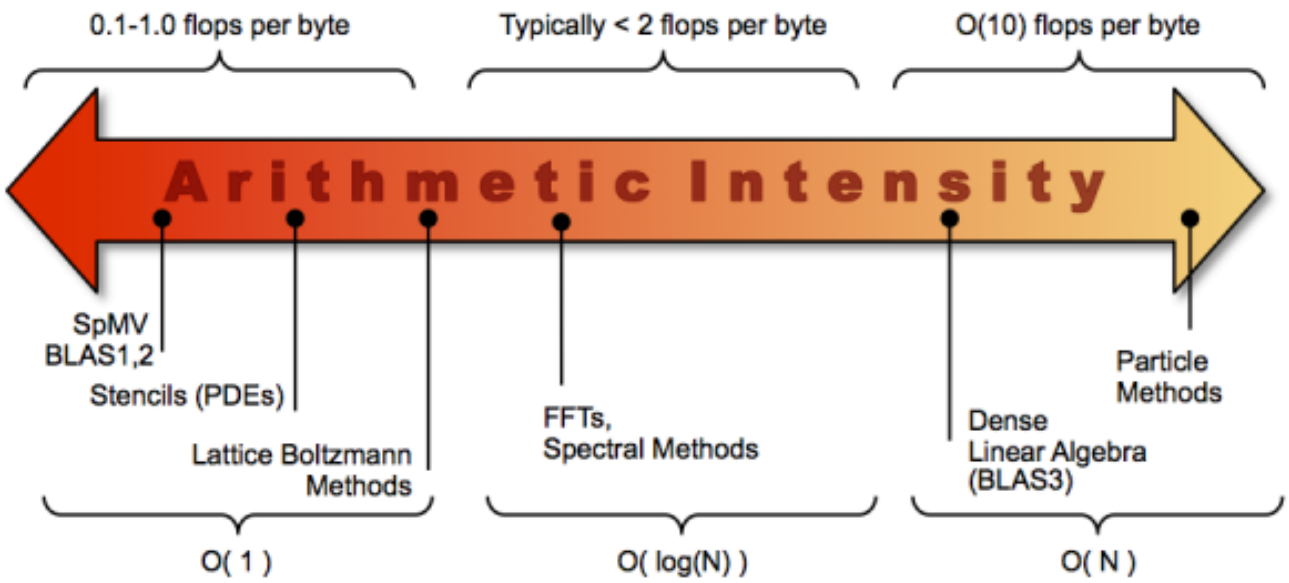
- Again, the computation is:

```
for (int i=0; i<n; i++)
  for (int j=0; j<n; j++)
    y[i] += A[i*n + j] * x[j];
```

- How many **FP ops / float** for our mat-vec multiplication?
  - y[i] += A[i*n + j] * x[j] each iteration with n * n iterations
  - FP ops = **2 * n * n** (one multiply and one add)
  - Float accesses = **n * n + 2n** (1 matrix and 2 vector accesses)
    - That's counting only cold misses but could be even more
  - So approx. **2 FP ops / float** (a far cry from 28 FP ops / float)

- In general, matrix-vector multiplication has low intensity
  → Fundamentally not suited for fast GPU computation

Operational Intensity on a Spectrum



* Courtesy of Lawrence Berkeley National Laboratory:
https://crd.lbl.gov/departments/computer-science/par/research/roofline/introduction/

- **BLAS**: Basic Linear Algebra Subprograms
  - BLAS 1: Vector operations only (e.g. saxpy) → Bad intensity
  - BLAS 2: **Ge**neral **M**atrix-**V**ector Multiplication (**GeMV**) → Bad intensity
  - BLAS 3: **Ge**neral **M**atrix **M**ultiplication (**GeMM**) → Good intensity

Matrix-Matrix Multiply: Good Arithmetic Intensity

- Matrix-multiplication:

```
for (int i=0; i<n; i++)
    for (int j=0; j<n; j++)
      for (int k=0; k<n; k++)
        C[i*n + j] += A[i*n + k] * B[k*n + j];
```

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \\ b_7 & b_8 & b_9 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}$$

- What's the arithmetic intensity for this program?
  - FP ops = **2 * n * n * n** (one multiply and one add)
  - Float accesses = **3 * n * n** (3 matrix accesses)
    - If we only have cold misses and no capacity misses
  - Arithmetic intensity = 2 * n / 3 = **0.66 * n** = **O(n)**
  - Implication: The larger the matrix size, the better suited for GPUs!
    - Important result for deep learning and other apps

Example: Computing $C(i,j) = A(i,k) * B(k,j)$

C program (on CPU)

```
void mm_cpu(float* C, float* A, float* B,
int n) {
  for (int i=0; i<n; i++)
    for (int j=0; j<n; j++)
      for (int k=0; k<n; k++)
        C[i*n + j] += A[i*n + k] * B[k*n + j];
}
```

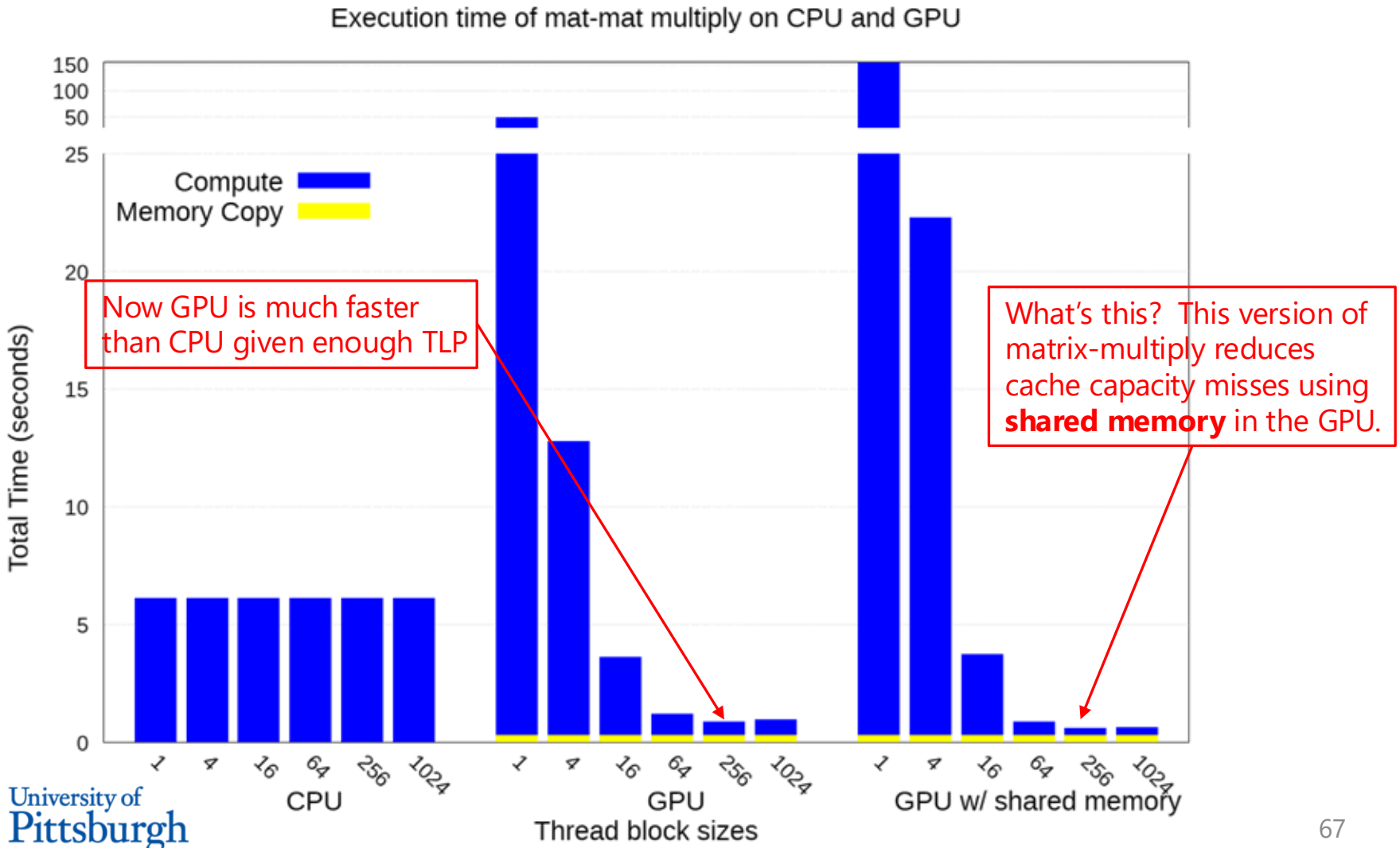
```
void main ()
{
  mm_cpu(C, A, B, n);
}
```

CUDA program (on CPU+GPU)

```
void mm_gpu(float* C, float* A, float* B, int n) {
  float Cvalue = 0;
  int i = blockIdx.y * blockDim.y + threadIdx.y;
  int j = blockIdx.x * blockDim.x + threadIdx.x;
  for (int k = 0; k < n; ++k)
    Cvalue += A[i * n + k] * B[k * n + j];
  C[i * n + j] = Cvalue;
}
```

```
void main ()
{
  dim3 dimBlock(block_size, block_size);
  dim3 dimGrid(n / dimBlock.x, n / dimBlock.y);
  mm_gpu <<<dimGrid, dimBlock>>> (C, A, B, n);
}
```

Performance Results for $C(i,j) = A(i,k) * B(k,j)$



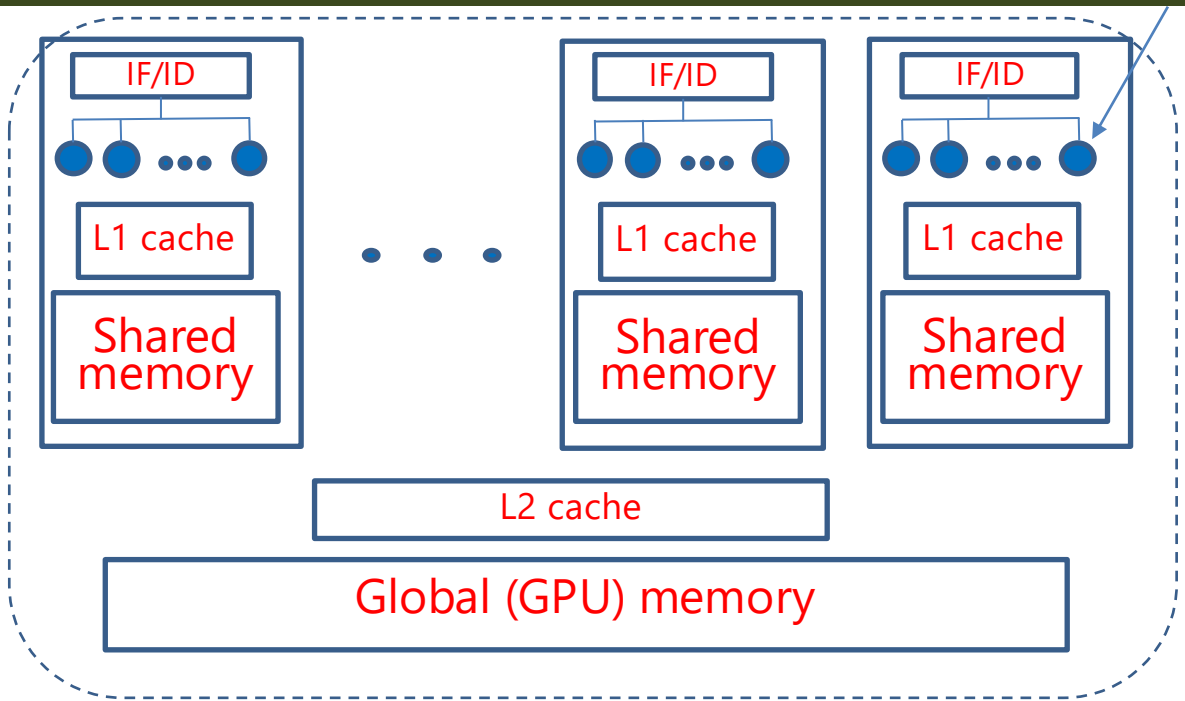
Capacity Misses can Reduce Arithmetic Intensity

```
for (int i=0; i<n; i++)
   for (int j=0; j<n; j++)
     for (int k=0; k<n; k++)
       C[i*n + j] += A[i*n + k] * B[k*n + j];
```

- The ideal arithmetic intensity for this program was:
  - FP ops = **2 * n * n * n** (one multiply and one add)
  - Float accesses = **3 * n * n** (3 matrix accesses)
  - Arithmetic intensity = 2 * n / 3 = **0.66 * n** = **O(n)**

- Only if we have **no capacity misses**. What if we did have them?
  - For B[k*n + j], reuse distance is the entire matrix of B
  - If B[k*n + j] always misses, memory accesses is in the order of **n^3**!

So what is Shared Memory?



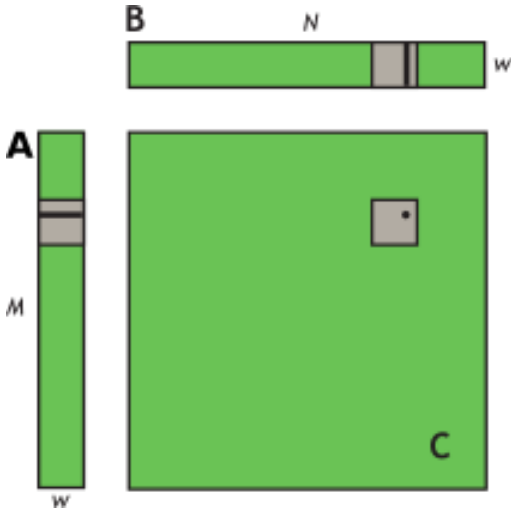
- **Shared Memory**: memory shared among threads in a thread block
  - Variables declared with **__shared__** modifier live in shared memory
  - Is same as L1 cache in terms of latency and bandwidth!
  - Storing frequently used data in shared memory can save on bandwidth

Loop Tiling with Shared Memory

- Store a "tile" within matrix in shared memory while operating on it
  - Can reduce accesses to DRAM memory

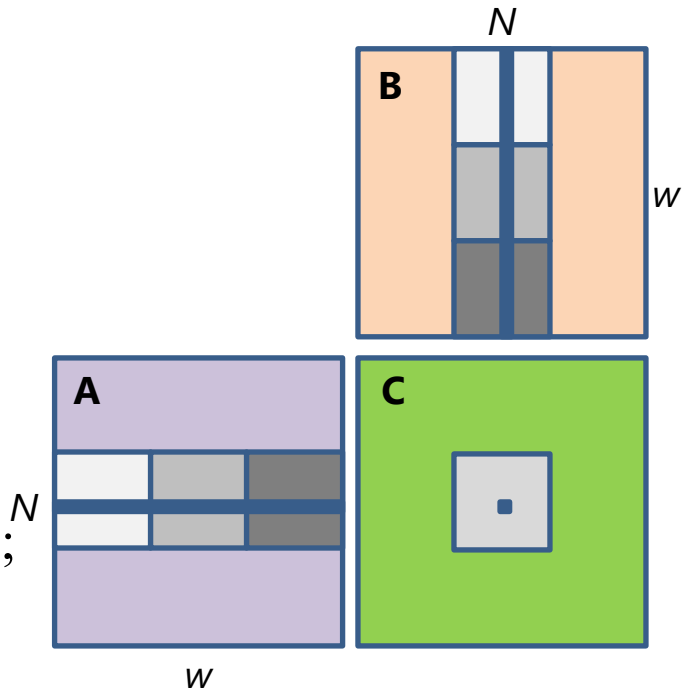
- Code in: *https://docs.nvidia.com/cuda/cuda-c-best-practices-guide/index.html#shared-memory-in-matrix-multiplication-c-ab*

```
__shared__ float aTile[TILE_DIM][TILE_DIM],
                 bTile[TILE_DIM][TILE_DIM];
int row = blockIdx.y * blockDim.y + threadIdx.y;
int col = blockIdx.x * blockDim.x + threadIdx.x;
float sum = 0.0f;
aTile[threadIdx.y][threadIdx.x] = a[row*TILE_DIM+threadIdx.x];
bTile[threadIdx.y][threadIdx.x] = b[threadIdx.y*N+col];
__syncthreads();
for (int i = 0; i < TILE_DIM; i++)
  sum += aTile[threadIdx.y][i]* bTile[i][threadIdx.x];
c[row*N+col] = sum;
```



- Assumption: TILE_DIM = w. What if w > TILE_DIM?

Loop Tiling with Shared Memory

- TILE_DIM is limited by amount of shared memory. What if w > TILE_DIM?
  - Now must load A and B tiles **w / TILE_DIM** times per thread block
- Now code will look like:

```
__shared__ float aTile[TILE_DIM][TILE_DIM],
                 bTile[TILE_DIM][TILE_DIM];
float sum = 0.0f;
for (int t = 0; t < w / TILE_DIM; t++) {
   …
   aTile[threadIdx.y][threadIdx.x] = …;
   bTile[threadIdx.y][threadIdx.x] = …;
   __syncthreads();
   for (int i = 0; i < TILE_DIM; i++)
      sum += aTile[threadIdx.y][i]* bTile[i][threadIdx.x];
   __syncthreads();
}
c[row*N+col] = sum;
```